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Propagation, dynamics and control of laminar premixed flames

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Abstract

Current state of knowledge of laminar premixed flame propagation, dynamics and control is reviewed. A distinction is made between planar and multi-dimensional flames. Based on the well-established theory of flame propagation of free adiabatic flames, different mechanisms to control planar flames, mainly based on heat transfer processes in porous media, but also using electrical and gravitational fields, are considered. Flame instabilities, leading to flame pulsations or oscillations, for instance in an acoustic field, are also considered. A simple physical model for the interaction between a premixed flame and a burner is presented to explain the main physical mechanisms of planar instabilities of burner-stabilized flames. Knowledge on flame propagation of planar flames is well-established and physical models to describe the behavior of these flames are mature, although it is expected that more sophisticated methods will be used in future research to control their behavior.

Multi-dimensional flames are also well understood, although the state of knowledge is less mature. Models for the interaction of flames with flow distortions, leading to flame stretch and curvature are derived. The influence of such distortions on the burning velocity and flame dynamics in terms of the Markstein number is studied. Weak-stretch models are available but multiple species chemistry and transport effects are less well understood, especially for rich and stoichiometric mixtures. Some ways to extend or improve current models are indicated. Current knowledge on the interaction of premixed (Bunsen-type) flames with flow distortions, such as acoustic waves, is also reviewed. Experimentally observed phenomena can be modeled satisfactory using detailed numerical models. However, for this research area, it is clear that current physical models or analytical descriptions are less mature because the origin of some phenomena is still largely unknown. Extensions of current (mostly G-equation like) models to account for flame–flow interaction, flame–flame interaction and flame–burner interaction are needed. To conclude: a multiple set of measures to adapt the flames to the combustion system of interest is at hand and different ways to influence the flame by heat transfer, acoustic waves and other less often used control mechanisms like electric fields are available. This is needed to modify and control premixed laminar flames in practical devices to guarantee safe and proper operation.
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1. Introduction

Laminar premixed flames constitute a central theme in combustion science in at least three respects. First, laminar premixed flames are used in many practical applications. Second, results obtained in laminar flame theory are often used in many other combustion studies, and finally, laminar premixed flames are very interesting from a fundamental scientific point of view. The close coupling between transport processes and chemical kinetics leads to a flame structure in the form of a thin layer propagating through the unburnt mixture. The burning velocity of the flame front is a key parameter in combustion. The burning velocity is very sensitive to details of kinetics and transport and is as such used extensively to develop and validate kinetic mechanisms and combustion models. Furthermore, laminar premixed flames display a broad spectrum of exciting phenomena, like pulsations and cellular structures, depending on the fuel, operating parameters and system considered.

Mallard and Le Chatellier [73] were the first researchers who studied the propagation of flames in premixed systems and the basis for modern combustion theory is set by Zeldovich and Frank-Kamenetski [137]. Matalon [79], Clavin [29], Sivashinsky [118] and Candel [22] previously reviewed flame dynamics of premixed flames. The key focus of the current review paper is to present a physical description of the propagation of premixed flames, which, depending on the flow and burner system, might lead to interesting flame dynamics. We also consider several methods to 'control' the flame propagation and dynamics, or in other words, to physically influence the flame such that it behaves as desired. We will focus on current state-of-the-art in modeling the physical behavior of premixed laminar flames. For that reason, we present knowledge in the form of theoretical descriptions, but we also consider numerical results and experiments, used to validate and confirm the knowledge obtained. Section 2 focusses on planar flames, while multi-dimensional effects are considered in Section 3.

2. Planar flames

Current state-of-the-art knowledge of the propagation of planar deflagaration waves in premixed mixtures will be reviewed in this section. The flow field considered will be a simple homogeneous plug flow rendering a completely one-dimensional problem. Effects of distortions of the flame and the flow in lateral directions largely influence flame behavior, but these effects will be discussed in the next section. Undistorted planar premixed flames in a plug flow mostly propagate with a constant velocity, referred to as the laminar burning velocity s_L, and such flames have been studied for a long time. It is difficult to study free laminar adiabatic premixed flames in the laboratory because such flames are hard to control. There are some methods to control premixed planar flames, i.e., use methods to influence the formation of premixed planar flames by 'external forcing'. This includes the application of well-balanced heat transfer processes, for instance using burners and porous media. Steady propagation of planar flames is not always possible, especially in case of flames inside a closed system. This then may lead to spontaneous pulsations or to oscillations in case of flames interacting with the acoustic field. Passive and active control forms a relatively new branch of research, sometimes used to influence these motions. Examples of active control may involve the application of external acoustic, electrical or magnetic fields or the adjustment of the gravitational field.

These are the topics of Section 2. First, in Section 2.1, we review the major structure of a premixed planar flame in a plug flow. The main physical behavior is considered in terms of a simple reaction kinetic mechanism. In reality, the structure is governed by detailed kinetics and transport processes involving many species. We will only discuss the major effects of detailed kinetics and transport on the flame structure. This is done in terms of an overall thin reaction layer combining even thinner layers, reviewed briefly in Section 2.2. The dynamics of planar flames stabilized on porous burners or inside porous inert media, as such influencing the dynamics and structure by external heat and mass transfer processes to control the flame propagation, will subsequently be considered in Section 2.3. Unsteady effects may arise and will be studied as well. This will be continued in Section 2.4, where the dynamics of flame oscillations in an external acoustic field will be reviewed. Other physical control mechanisms, based on external (electrical and gravitational) fields are used more and more. A first glance on such methods will be given in the final Section 2.5.

2.1. Structure and propagation of one-dimensional premixed flames

The propagation velocity of premixed flames has been studied since more than a century. The essential result that the burning velocity is proportional to the square root of the reaction rate ω and the square root of λ/c_p, with λ the conductivity and c_p the specific heat, has been recognized since first studies in this area, see e.g., Mallard and Le Chatelier [73]. This already indicates the importance of both reaction kinetics as well as transport processes. The coupling between kinetics and transport leads to a flame front, a thin layer, separating the unburnt and burnt gases. The

hydrodynamic length scale associated with the geometry of the combustion system is much larger. This flame front consists of two sub-layers. In the first, the unburnt mixture is not reacting but is heated by conduction and this zone is referred to as the diffusive (preheating) layer and has thickness $\delta_T = \lambda/(\rho c_p s_L)$. Near the burnt side of the flame front an even thinner reaction layer with thickness δ_r is embedded, where chemical reactions take place. As an example, the global structure of a lean methane–air flame is presented in Fig. 1, but other hydrocarbon flames behave similarly. The separation of scales, i.e., $\ell_f \gg \delta_T \gg \delta_r$, has been used to develop theoretical models; see [134] for an overview.

Fundamental theoretical models for describing the structure and propagation of laminar premixed flames have been established by Zeldovich and Frank-Kamenetski in the 1930s [137] (see [136] for a more recent synthesis). This was later cast in the framework of matched asymptotic expansions by Bush and Fendell [21]. Activation energy asymptotics (AEA) inspired by the work of Zeldovich was extensively explored by Williams [132], Buckmaster and Ludford [19] and many others. In most of these early papers, the propagation of a one-dimensional adiabatic flame is studied by considering a one-step irreversible reaction. The transport equations for the lean reactant mass fraction Y and the temperature T can then be written as

$$m\frac{\partial Y}{\partial x} - \frac{1}{\mathrm{Le}}\frac{\lambda}{c_p}\frac{\partial^2 Y}{\partial x^2} = -\omega, \tag{2.1a}$$

$$m\frac{\partial T}{\partial x} - \frac{\lambda}{c_p}\frac{\partial^2 T}{\partial x^2} = \frac{Q}{c_p}\omega, \tag{2.1b}$$

where $m = \rho s_L$ is the (constant) mass burning rate, i.e., it is the mass flux relative to the flame and

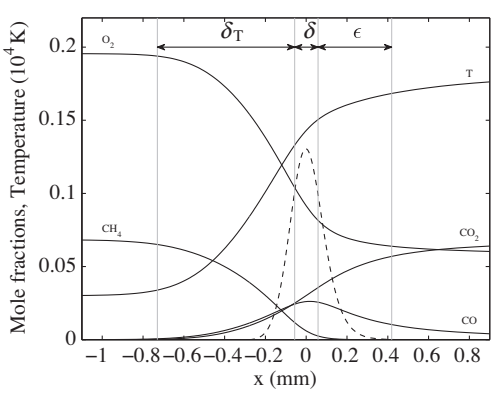

Fig. 1. Structure of a lean methane/air flame (equivalence ratio $\Phi = 0.6$), including definition of different layers: inert preheating layer δ_T, reaction layer δ_r consisting of an inner layer with thickness δ and oxidation layer with thickness ϵ; dotted line indicates the heat release profile.

$\mathrm{Le} = \lambda/\rho D c_p$ the Lewis number indicating the ratio of thermal diffusivity λ/c_p and mass diffusivity ρD. Furthermore,

$$\omega(Y,T) = \rho Y A\, \mathrm{e}^{-T_a/T}, \tag{2.2}$$

is the reaction rate, Q the heat release and $T_a = E_a/R$ the activation temperature. The boundary conditions for (2.1) read $T = T_u$, $Y = Y_u$ for $x \to -\infty$ and $T = T_{ad} = T_u + QY_u/c_p$, $Y = 0$ for $x \to \infty$. Combining the two transport equations and assuming an infinitely thin reaction layer, equivalent to $T_a \to \infty$, the following approximation for the mass burning rate m_0 was derived by Zeldovich and Frank-Kamenetski [137]:

$$m_0 = \sqrt{2\frac{\lambda}{c_p}A\rho_b}\,\frac{1}{\mathrm{Ze}}\,\mathrm{e}^{-T_a/(2T_{ad})}, \tag{2.3}$$

where the subscript '0' indicates that this is the undistorted adiabatic mass burning rate. For now, we consider adiabatic flames; while external heat losses are considered furtheron. Furthermore, $\mathrm{Ze} = T_a(T_{ad} - T_u)/T_{ad}^2$ is the Zeldovich number, which is a dimensionless activation temperature; typically $\mathrm{Ze} = \mathcal{O}(10)$. To derive (2.3), it was assumed that $\mathrm{Le} = 1$; a detailed analysis for $\mathrm{Le} \neq 1$ and generalizations of (2.3) for higher order reactions can be found in [136].

A different approach to compute the mass burning rate is based on matched asymptotic expansions (MAE) [21]. Applying MAE, Bush and Fendell also found the expression (2.3) for the mass burning rate. A higher-order (in 1/Ze) and more accurate result is due to Williams [132]. Yet another method to determine m_0 is the integral analysis (IA) developed by Law [70].

2.2. Multi-layer structure of one-dimensional flames

While simplicity is the elegance of the previous analysis, it carries the drawback that it is not based on realistic chemical kinetic mechanisms. Differences between the burning velocities in different mixtures are governed by differences in kinetics and transport processes. Numerical simulation of the one-dimensional governing transport equations is possible nowadays even for large mechanisms with detailed kinetics and transport models. Numerical data of laminar burning velocities reproduce experimental results quite well (within roughly 10%) for lower hydrocarbon–air mixtures. This accuracy is not surprising since s_L is one of the key parameters used to fit the rate parameters in the mechanisms. Furthermore, experimental data for laminar burning velocities in a large variety of mixtures have been obtained using different techniques (spherically propagating flames in vessels, planar stagnation flames and planar adiabatic flames stabilized on burners).

Accuracy of current experimental data amounts to about 1 cm/s, depending on the situation. The link between measured and numerical burning velocity data for different fuels and analytical results of the previous section remains obscure due to the sensitivity to kinetics and transport processes.

Peters, Williams, Seshadri and others [98, 112,113] have shown in the last decades that it is possible to use the theoretical framework of MAE also for flames with more detailed kinetic mechanisms. For the case of stoichiometric methane–air flames, they started from a detailed mechanism and systematically reduced the mechanisms first to four steps and finally to two steps. The two-step mechanism enables the use of scale separation again by introducing two reaction layers instead of one: a very thin inner layer with thickness δ where CO and H_2 are produced from the fuel and a thicker oxidation layer with thickness ϵ where CO and H_2 are oxidized (see Fig. 1). This involves the matching of four zones: $\ell_f \gg \delta_T \gg \epsilon \gg \delta$. The final result for the burning velocity is given by [98]:

$$s_L^2 = C\left(\frac{T_u}{T_0}\right)^2\left(\frac{T_{ad}-T_0}{T_{ad}-T_u}\right)^4, \tag{2.4}$$

where the temperature in the inner layer T_0 is computed from the matching procedure and is given by

$$\frac{RT_0k_1^2(T_0)}{k_{11}(T_0)k_5(T_0)} = 1.5\,p\frac{\mathrm{Le_F}}{\mathrm{Le_{O_2}}}. \tag{2.5}$$

The parameters in Eq. (2.5) are the Lewis numbers and rate parameters of important reactions in the detailed mechanism. The factor C also contains many rate parameters and Lewis numbers. Note that there is no explicit dependence on the activation temperature in these equations. However, an effective activation temperature can be derived from:

$$T_a = -\frac{\mathrm{d}}{\mathrm{d}(1/T_{ad})}\left(\ln s_L^2\right) \approx \frac{4T_{ad}}{T_{ad}-T_0}. \tag{2.6}$$

Although this model qualitatively predicts the structure and propagation of a stoichiometric methane–air flame, it was not able to reproduce measurements and numerical data quantitatively. For this reason, a number of extensions and improvements have been introduced over the years. In [112] for instance, Seshadri and Peters introduced so-called fuel consumption and radical consumption layers inside the inner layer to improve accuracy. Extensions to lean and rich methane–air flames and flames of some other mixtures have been introduced as well. Seshadri reviewed the use of asymptotic methods to model flame structure and propagation in different mixtures in [113].

2.3. Impact of heat transfer

It is difficult to control free planar flames in the laboratory. The control of such flames, which is for instance important for measuring the adiabatic burning velocity, has been subject of study for many years. Perturbations in the flow (e.g., planar stagnation flames [67]), the flame geometry (e.g., spherically expanding flames [44]) and the heat balance (e.g., burner stabilized flames and heat-flux burner stabilized flames [72]) have been used to control their motion to determine the adiabatic burning velocity. In all these situations, except for the heat-flux method, extrapolations to the unperturbed planar adiabatic state have to be performed to determine the laminar burning velocity, because the unperturbed flames are unstable. In this subsection, we will consider the control of planar flames by heat transfer in such a way that they are stabilized on top of or inside porous inert material (PIM). Different materials are used for stabilizing planar flames in practical systems, such as porous ceramics, sintered and fibre metals, but also perforated ceramics and metals. Flame temperature is not adiabatic anymore if heat losses are non-zero: $T_b = T_{ad} - q/c_p$, where T_b is the non-adiabatic flame temperature and q the total heat loss. We focus on convective and conductive heat transfer, while radiative transfer might have an additional effect, although this effect is mostly negligible for small-scale flames.

The combustion and stabilization of flames interacting with PIM has been studied extensively. A central result of these studies is the so-called stabilization diagram, first introduced by Sathe et al. [115]. This diagram is show in Fig. 2 and indicates how the burning velocity of the flame structure behaves as a function of the position inside the PIM. For low v_u, there are three 'equilibrium' points where the gas flow velocity can become equal to the local burning velocity. Two points, one near the upstream boundary of the material and one outside the burner near the downstream boundary of the PIM are mostly stable points, while the point inside the PIM near the outflow boundary is unstable. The two stable points indicate two operating modes, already mentioned by Viskanta [127]: flames stabilized inside and on top of PIM. Transitions between the stable states are possible for some materials and such situations are studied for instance by Lammers and de Goey [64]. These phenomena are referred to as flash-back or blow-off. Figure 3 is a zoom of Fig. 2 and shows the opening of a flash-back window between 18 and 22 cm/sif the surrounding temperature is increased. We will consider both operating modes, combustion on top of and inside the PIM, in the following.

Flames on top of PIM have been studied extensively in the past. Laboratory burners, like McK-

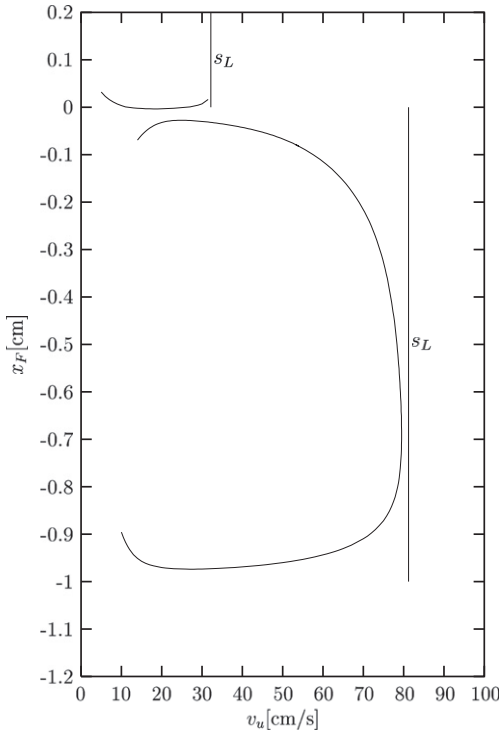

Fig. 2. Stabilization diagram for methane–air flames on top and inside PIM in terms of stabilization velocity v_u versus the position x_F close to or within the PIM. The material extends from $x_F = -1$ cm to $x_F = 0$ cm.

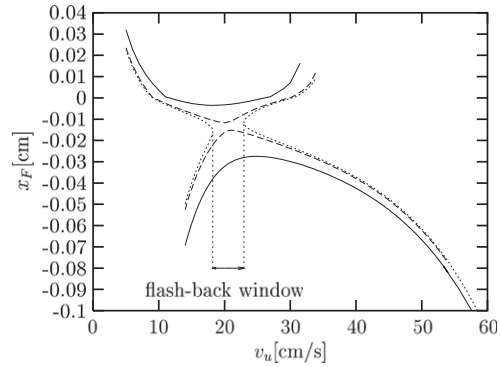

Fig. 3. Zoom of Fig. 2, close to the burner outlet (drawn line). If the surrounding temperature increases, stabilization lines approach (dashed line) and eventually cross leaving a flash-back window (dotted line).

enna products burners [89] and heat flux burners [72] are academic burners which use the heat loss of planar premixed flames to stabilize them. These burners are often used to study the structure of such flames, for instance using laser-diagnostic techniques like Laser-Induced Fluorescence (LIF). Surface stabilized flames are also widely used in practical combustion systems nowadays, especially in small-scale house-hold burners and boilers because of their wide operating range in terms of equivalence ratios and specific powers available to stabilize the flames, their high radiative power and their low NO emissions. Heat is transferred from the gas to the burner material in the preheating zone of the flame, where chemical effects are still unimportant. This heat is finally transferred to the environment by cooling the burner material externally or by radiation (e.g., for ceramic burners), which explains the lower flame temperatures and low NO emissions.

It has been shown by de Goey et al. [39] that surface stabilized flames behave exactly the same as adiabatic planar flames propagating in a mixture at a lower initial temperature $T'_u = T_u - q/c_p$ as long as the flame temperatures $T_b = T_{ad} - q/c_p$ q being the heat transferred to the burner), of both flames are the same and as long as the heat transfer of the flame to the burner only takes place in the preheating zone of the flame structure, where $\omega = 0$. For the mass burning rate m of a non-adiabatic flame with flame temperature $T_b < T_{ad}$ we find

$$m = \frac{1}{c_p(T_b - T'_u)}\sqrt{2Q\lambda\int_{T'_u}^{T_b}\omega\,dT} = \frac{1}{c_p(T_{ad} - T_u)}\sqrt{2Q\lambda\int_{T_u}^{T_b}\omega\,dT}. \quad (2.7)$$

As a result, it can be shown that the mass burning rate m and the flame temperature T_b are effectively related to each other by

$$\frac{m}{m_0} = \exp\left(-\frac{1}{2}T_a\left(\frac{1}{T_b} - \frac{1}{T_{ad}}\right)\right). \quad (2.8)$$

This relation has already been found experimentally by Kaskan in 1967 [57] and has been verified by many others afterwards. The heat loss profile of the flame has no influence on the mass burning rate as long as it is concentrated in the preheating zone of the flame, as shown by de Goey et al. [39]. This observation has been used by de Goey and his group to develop the so-called heat-flux method for stabilizing adiabatic flames and for measuring the adiabatic mass burning rate of gas mixtures in a very accurate way (see for instance [13,48,58,72]). The heat-transfer profile in this situation consists of a gain part in a cold part of the flame and a loss part in a somewhat hotter region near the outflow of the burner material. The integrated heat transfer is zero, inducing an adiabatic situation and the possibility to stabilize adiabatic flames.

Flames propagating inside PIM have been studied extensively as well (see [125,50] for recent reviews). The flame structure changes dramatically compared to the free flame due to the fact

that the heat transfer profile is extended over the complete flame structure and now interacts with the reaction layer. Generally, it can be stated that heat is gained by the gas from the PIM in the cold part while heat loss dominates the flame in the burnt gas zone. This leads to so-called super-adiabatic or excess-enthalpy flames, where flame temperatures are locally higher than the adiabatic value in the reaction layer even if the complete system is adiabatic [45]. Burning velocities are rising considerably compared to m_0 due to two effects: the higher flame temperature $T_b > T_{ad}$ and the higher effective conductivity $\lambda_{eff} > \lambda$ to preheat the mixture in the preheating zone (see Eq. (2.7) where λ and T_b become higher). In practice, the two PIM boundaries radiate heat to the environment, thereby leading to heat loss and enabling flame stabilization at lower gas velocities. If the heat transfer is not well established or controlled, flash-back (ignition of the mixture in the upstream part) might occur. To prevent this from happening, additional layers of materials have been added to the upstream part of the PIM in practical systems (see for instance [47]) as flame arrestor and/or flow distributor.

As mentioned, the 'equilibrium' point near the burner outflow in the stabilization diagram is not stable, forcing the flame to move upstream or downstream to one of the other two 'equilibrium' points. This does not mean that these two 'equilibrium' points are always stable. Both operating modes have shown pulsating solutions. Blackshear et al. [10] and El-Hamdi et al. [34] observed a variety of pulsating instabilities of flames stabilized on top of a perforated burner, in the form of axial, radial, drum-head and spiral modes. As an example, see Fig. 4. For the analysis of the axial pulsation a nonlinear model is required if we are interested in, for example, the flame oscillation amplitude [80,81], while a linear model is sufficient to study the onset of pulsating instabilities. In analyzing the stability of flames use is often made of the constant density (or thermal diffusive) model [20,117], so that the mass and momentum equations are decoupled from the species and energy balance equations. Because of the decoupling, the pulsating flame problem becomes analytically tractable. However, this is at the expense of the less realistic constant density assumption. For one-dimensional flames the constant density approximation is not needed, because the Von Mises transformation allows the density and velocity perturbations to be eliminated. Using the Von Mises transformation, Margolis [74] and McIntosh and Clarke [84] derived a dispersion relation using large activation energy asymptotics, which determines under what conditions a flame becomes unstable. These models predict that thermo-diffusive effects might lead to pulsations for Le > 1. An explanation that is sometimes used for this, is that the enthalpy within the preheating zone due

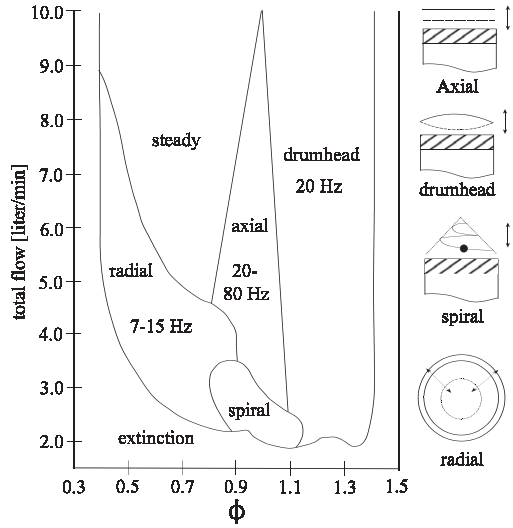

Fig. 4. State diagram showing the various modes of instability occurring in planar burner-stabilized methane/air flames for low flow rates at various equivalence ratios and at a pressure of 1/3 atm. Note that 10 l/min indicates a velocity of approximately 5–6 cm/s [35].

to Le > 1 processes is higher than the value of the flame front. A flame moving in the direction of the preheating zone due to some distortion, will then be accelerated due to this surplus of enthalpy, leading to instability. This is a rather simple explanation and it is unclear for which cases this explanation holds. Theoretically it is shown that pulsations are difficult to obtain in adiabatic flames because the Lewis number has to be very large for this to happen [51,117]. To increase Lewis numbers, Pearlman and Ronney used helium instead of nitrogen and observed pulsations in near adiabatic C_4H_{10}–O_2–He mixtures [97]. On the other hand, in free flames that are susceptible to a volumetric heat loss (e.g., by radiation) or in burner stabilized flames, the heat loss promotes oscillatory instability [20,51]. In burner stabilized flames oscillations may even persist if the Lewis number is 1 if velocities are sufficiently low [53,74,84]. Since the Lewis numbers of the major species in methane/air combustion do not deviate much from 1, pulsations are likely to occur in such flames.

All these phenomena are mainly governed by the interaction of the heat transfer to the burner with the flame propagation and are especially manifest for low velocities and large heat losses, near the extinction limits. To grasp some of the physics of this process, the following simple model shows the basics of these phenomena. When a flame oscillates on top of a burner, the heat loss at the burner oscillates also, because of changes in the temperature gradient at the burner. This results in an enthalpy wave propagating from the burner to the reaction layer and flame dynamics is influenced due to the corresponding flame

temperature changes and the time lag involved. Consider a reaction sheet stabilized at $x = x_f$ above a 'perfect' burner at $x = 0$ having a constant temperature $T = T_u$. If the flame is steady, the mass flow rate $\rho_u \bar{u}_u$ equals the mass burning rate of the flame sheet $\bar{m}$, where the bar above a variable denotes its mean value. In case of a temporal variation in the flow rate $u_u(t) = \bar{u}_u + u'_u(t)$, the flame starts moving and m becomes a function of the space coordinate x and time t. In our model, the flame is treated as a 'rigidly' oscillating structure only moving without internal changes, i.e., $m = m(t)$, implicitly assuming that 'internal flame stretching' $\rho K = \partial m/\partial x$, can be neglected [106]. The question now is: how does the corresponding variation in the mass burning rate $m(t)$ relate to the mass flow rate changes $\rho_u u_u(t)$? If we assume that a change in flame temperature T_b leads to an instantaneous change in m according to Eq. (2.8), we find

$$\frac{\partial m}{\partial t} = \frac{\mathrm{Ze}}{2} \frac{\rho_u \bar{u}_u}{\overline{T}_b - T_u} \frac{\partial T_b}{\partial t} = \frac{\mathrm{Ze}}{2} \frac{\rho_u \bar{u}_u}{c_p(\overline{T}_b - T_u)} \frac{\partial H}{\partial t}(\psi_f, t), \tag{2.9}$$

where $\mathrm{d}H = Q\,\mathrm{d}Y + c_p\,\mathrm{d}T$ is the fluctuation in the 'total enthalpy' of the system and depends not only on the time t, but also on the (von-Mises) space coordinate ψ defined by $\mathrm{d}\psi = \rho\,\mathrm{d}x/\rho_u$. In the derivation of (2.9) we assumed that $Y_b = 0$, i.e., fuel is depleted in the flame front at $\psi = \psi_f$. Fluctuations in H are transported with the gas velocity $\bar{u}_u$, so that heat losses to the burner $\psi = 0$ are detected at the flame front with a time lag $\psi_f/\bar{u}_u$. In general, the shape of the enthalpy wave changes due to damping effects, due to molecular diffusion when convective wavelengths are comparable to diffusion length scales (flame thickness). If damping is neglected, and assuming $\mathrm{Le} = 1$, we may write:

$$\frac{\partial H}{\partial t}(\psi_f, t) = \frac{\partial H}{\partial t}(0, t - \psi_f/\bar{u}_u) = Q\frac{\partial Y}{\partial t}(0, t - \psi_f/\bar{u}_u), \tag{2.10}$$

where we used that $T = T_u$ is constant at the burner outflow. Using the kinematic equation for the flame motion in terms of Y:

$$\rho_u \frac{\partial Y}{\partial t}(\psi, t) = (m(t) - \rho_u u_u(t))\frac{\partial Y}{\partial \psi}(\psi, t), \tag{2.11}$$

we finally find a first-order differential-delay equation for $m(t)$:

$$\frac{\partial m(t)}{\partial t} + \frac{Z}{\tau_f} m(t') = \frac{Z}{\tau_f}\rho_u u_u(t'), \quad t' = t - \tau_d, \tag{2.12}$$

where we introduced the delay time $\tau_d = \psi_f/\bar{u}_u$, the 'flame time' $\tau_f = \delta_T/\bar{u}_u$ and the 'feed-back' coefficient Z given by

$$Z = \frac{\mathrm{Ze}}{2}\frac{T_{ad} - \overline{T}_b}{\overline{T}_b - T_u}. \tag{2.13}$$

Equation (2.12) describes the motion of the flame due to velocity fluctuations $u_u(t')$ and will also be used in the next subsection to describe the acoustic transfer function governing the interaction of acoustic waves with premixed planar burner-stabilized flames. However, also if no acoustic waves were present ($u_u = \bar{u}_u$), this equation predicts pulsating solutions, especially if Z is large.

For a harmonic forcing term $u_u(t) = \bar{u}_u + \hat{u}_u \exp(\mathrm{i}\omega\tau)$, the mass flow rate m behaves harmonically, while $\bar{m} = \rho_u \bar{u}_u$, and the complete solution of Eq. (2.12) is given by

$$m(t) = \bar{m} + \hat{m}\,\mathrm{e}^{\mathrm{i}\omega t} - \hat{m}\,\mathrm{e}^{\alpha t/\tau_d}\,\mathrm{e}^{\mathrm{i}\beta t/\tau_d}, \tag{2.14}$$

where we used the initial condition $m(0) = \bar{m}$. Furthermore, α and β have to satisfy the relations

$$\alpha = -\widehat{Z}\,\mathrm{e}^{-\alpha}\cos(\beta), \quad \beta = \widehat{Z}\,\mathrm{e}^{-\alpha}\sin(\beta), \tag{2.15}$$

where $\widehat{Z} = \psi_f Z/\delta_T$.

The homogeneous part of the solution (2.14) indicates when solutions are stable or unstable. Stable solutions are found when $\alpha < 0$, while the solution becomes unstable if $\alpha > 0$, even if there is no external forcing ($\hat{u}_u = 0$). Unstable solutions are found when $\alpha > 0$, for which $\pi/2 < \beta < \pi$ and $\widehat{Z} > \pi/2$. In Fig. 5 an example is shown for a burner-stabilized methane/air flame. Flame temperature $\overline{T}_b$, stand-off distance given by

$$\frac{\psi_f}{\delta_T} = \ln(T_{ad} - T_u)/(T_{ad} - \overline{T}_b) \tag{2.16}$$

and $\widehat{Z}$ are plotted as function of $\bar{u}_u$. Note that $\widehat{Z} > \pi/2$ for velocities lower than approximately 15 cm/s, indicating instability for velocities $\bar{u} < 15$ cm/s [74]. In numerical (and experimental) observations, instability is found for much lower velocities. This indicates that this theory is insufficiently accurate to predict instability. If damping of enthalpy waves is considered, a lower value for the critical velocity is found, in better accordance with experiments and other theoretical models [105]. Qualitative agreement exists between models, numerical results and experiments. However, quantitative agreement is not accomplished yet due to the sensitivity of the models to small changes, especially if one relies on simplified kinetics, as shown by Gielen [38]. Flame pulsations are also observed for flames inside PIM's [64] although they have not been studied in a systematic way.

2.4. Planar burner stabilized flames in an acoustic field

Combustion in closed systems like gas turbine chambers or central heating boilers often suffers from flow and flame oscillations. This is most

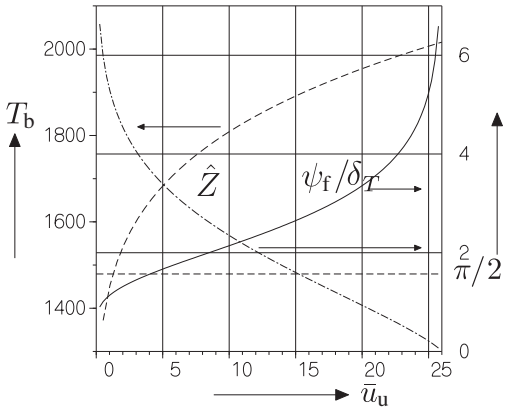

Fig. 5. Feedback coefficient $\widehat{Z}$, scaled stand-off distance ψ_f/δ_T and flame temperature T_b as function of $\bar{u}_u$ for a methane/air flame. Parameter values are: $\Phi = 0.8$, $T_{ad} = 2016\,K$, $T_u = 300\,K$, $T_a = 34.566\,K$ and $s_L = 26\,cm/s$.

often not desired because this may lead to severe noise emission or may be detrimental for structural integrity. Such thermo-acoustic oscillations have been studied for a very long time. Early examples are the 'singing flame' reported in 1777 by Higgins [49], the Sondhauss tube [119] and the Rijke tube [104]. Raun et al. [102] presented an extensive literature overview. Lord Rayleigh already showed in 1878 [103] that acoustic energy is produced by an oscillating heat source if it is in phase with the pressure oscillations in the system. This occurs for instance when acoustic waves in a combustion system interact with the flame. The velocity fluctuation u' of an acoustic wave, on the other hand, might induce the above-mentioned fluctuating heat release rate q' of the flame in several ways. In non-premixed flames, for instance, equivalence ratio variations might be the driving force of this phenomenon. In free premixed (Bunsen-type) flames, flame surface area oscillations are (at least partly) responsible. This system is analyzed in Section 3. Searby recently performed very nice experiments on the interaction of free planar flames in a tube with acoustic waves (see e.g., [111]). In case of planar burner-stabilized flames, heat loss fluctuations to the burner due to the flame motion relative to the burner are the cause of heat release fluctuations. This will be considered in this subsection. The problem is for instance relevant for central heating systems in which surface-stabilized flames are applied and in other combustors using surface-stabilized burners.

We consider the most important case of planar acoustic waves traveling in the direction of a relatively long duct for which the cross-flow dimensions are much smaller. This enables the use of the transfer matrix method [90] to describe the acoustic behavior of the system, in which each acoustic element is described by a single matrix coupling the acoustic field on both sides of the element. The acoustic behavior of the system is determined by the eigen modes of the characteristic equation governed by the system matrix. Furthermore, in the low-Mach number limit, the acoustic wavelengths become infinitely large compared to the length scales in the flame and the flame can be regarded as compact. The flame response can be evaluated assuming incompressible flow on either side of the flame and connected by jump conditions. For this case the flame transfer matrix for the acoustic field is given by [90]:

$$\begin{pmatrix} p'_b \\ u'_b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & T_{22} \end{pmatrix} \begin{pmatrix} p'_u \\ u'_u \end{pmatrix}, \tag{2.17}$$

where the subscripts u and b refer to the unburnt and burnt sides of the flame, respectively. Due to the low Mach numbers, the pressure waves are not coupled to the flow and the $T_{22} = u'_b/u'_u$ matrix element, which couples the velocity fluctuations at both sides of the system, is the only remaining function of interest.

The transfer matrix element $T_{22} = u'_b/u'_u$ is often difficult to evaluate experimentally and it is therefore more convenient to use the so-called flame Transfer Function (TF), being the ratio of the heat release fluctuation q' to the velocity excitations u'_u, scaled with their mean values $\bar{q}$ and $\bar{u}_u$, i.e.,

$$\mathrm{TF}(\omega) = \frac{q'(\omega)/\bar{q}}{u'_u(\omega)/\bar{u}_u}, \tag{2.18}$$

which is defined here as a complex function in frequency domain $\omega = 2\pi f$.

A typical experimental setup to measure the TF includes a burner with flame, an external forcing unit and a means to measure flow velocity oscillations u'_u and the (volume integrated) heat release rate response q'. As an appropriate indicator for the flame heat release rate the chemiluminescence of OH^* or CH^* radicals are typically used. The element T_{22} is found from the flame TF by integrating the heat release fluctuations q' derived from the TF function over the domain using the energy conservation equation. However, in order to predict the heat release fluctuations q' of such flames, it is necessary to model the dynamics of the complete flame structure in the perturbed flow.

Planar burner-stabilized flames in an acoustic field have been theoretically studies extensively, e.g., by McIntosh et al. [84–86], van Harten et al. [46] and Buckmaster [20]. Using matched asymptotic expansions, analytical expressions for T_{22} have been derived by McIntosh in a number of subsequent publications, see for instance [85,86]. For the case of unit-Lewis numbers the result can be found in [87,102]:

$$T_{22} = 1 + \frac{1}{2}\frac{\overline{T}_b - T_u}{T_u} \times \frac{\left(1+\sqrt{1+4i\hat{\omega}}\right)\exp\left(-\frac{1}{2}\frac{\psi_f}{\delta_T}\left(1+\sqrt{1+4\,i\hat{\omega}}\right)\right)}{\sqrt{1+4i\hat{\omega}}\exp\left(-\frac{\psi_f}{\delta_T}\sqrt{1+4i\hat{\omega}}\right)+\frac{i\hat{\omega}}{Ze}}, \tag{2.19}$$

where $\hat{\omega} = \omega\tau_f$.

The systematic study of McIntosh shows the presence of strong amplification of velocity perturbations in T_{22}, mostly in the low frequency range, with a shape very much like a resonance. However, the physical background of this phenomenon is hard to grasp from his analysis. For this reason, we used the simple physical model of a rigid flame oscillating on top of a burner to explain this behavior, introduced in the previous subsection. As mentioned before, our model is restricted to unit Lewis numbers (as Eq. (2.19)) and assumes no damping of enthalpy waves. The mass burning rate of the flame $m(t) = \bar{m} + \hat{m}\exp(i\omega t)$ due to 'forcing' by an acoustic wave can be found from (2.12) assuming that there are no self-sustained oscillations ($\alpha < 0$), so that the homogeneous part of the solution damps out and the particular solution is preserved. This relates the amplitude $\hat{m}$ of the mass burning rate fluctuations to that of the acoustic forcing $\hat{\phi}$, i.e., $\hat{m} = \mathscr{A}\hat{\phi}$, with

$$\mathscr{A}(\hat{\omega}) = \left(1 + \frac{i\hat{\omega}}{Z}\,e^{i\hat{\omega}\psi_f/\delta_T}\right)^{-1}. \tag{2.20}$$

Here, $\hat{\omega}\psi_f/\delta_T$ is the phase shift of the enthalpy wave between burner and flame front due to the time lag. Rook et al. [106] did take into account damping of enthalpy waves, still neglecting 'internal flame stretching' and obtained

$$\mathscr{A}(\hat{\omega}) = \left(1 + \frac{i\hat{\omega}}{Z}\exp\left(-\frac{\bar{\psi}_f}{\delta_T}\frac{1}{2}\left(1-\sqrt{1+4i\hat{\omega}}\right)\right)\right)^{-1}. \tag{2.21}$$

Note that (2.21) reduces to (2.20) if $\hat{\omega}$ is small, i.e., for long wavelengths, when damping is negligible. By using energy conservation T_{22} can be derived from Eq. (2.21), leading to [106]:

$$T_{22} = 1 + \frac{\overline{T}_b - T_u}{T_u}\mathscr{A}(\hat{\omega}) + \frac{T_{ad} - \overline{T}_u}{T_u} \times \frac{1-\mathscr{A}(\hat{\omega})}{i\hat{\omega}}\frac{1}{2}\left(1+\sqrt{1+4i\hat{\omega}}\right). \tag{2.22}$$

The simple model presented here is valid for lower frequencies, while at higher frequencies damping effects start playing a role as well as 'internal flame stretch' [105]. Nevertheless, the simple model does capture all other physical phenomena of the system. For instance, Eqs. ((2.20)–(2.22)) do display the resonance behavior mentioned above.

The resonance phenomenon was already observed by McIntosh et al. [83]. In the simple model presented here, this phenomenon is easily recognized when the phase lag in the enthalpy wave between burner and flame front $\hat{\omega}\psi_f/\delta_T$ equals $\pi/2$ in (2.20) (for a graphical representation, see [106]). In case of a $\pi/2$ phase lag in the enthalpy wave when it travels with velocity $\bar{u}_u$ from the burner to the flame front, minimal heat loss at the burner is 'felt' by the reaction front at the moment when the flame moves with highest velocity towards the burner. This causes high amplitudes in the flame velocity fluctuations. The resonance is clearly visible in Figs. 6 and 7, which present gain and phase of T_{22} as function of frequency for different models. The model of McIntosh (Le = 1), Eq. (2.19), is compared with the model of Rook, Eq. (2.22), and numerical results computed from the complete set of conservation laws, using one-step and multi-step kinetics.

Experimental results are also in line with the behavior presented in Figs. 6 and 7. However, it appears that the behavior is quite sensitive to the properties of the burner plate (like material, conductivity, emissivity and porosity). To predict what happens with realistic burners, Schreel et al. [116] studied the influence of burner and flow parameters on flame acoustics. Experiment and numerical result are well in agreement, but analytical models need some adaptation. To improve the agreement, the simple model was extended by taking burner material properties into consideration, but assuming perfect heat transfer between the gas and solid. The surface temperature T_{surf} of the burner at outflow ($\psi = 0$) now increases above T_u, the stand-off distance ψ_f decreases and T_{surf} enters as an additional parameter in the equations. The value of T_{surf} depends on the heat losses

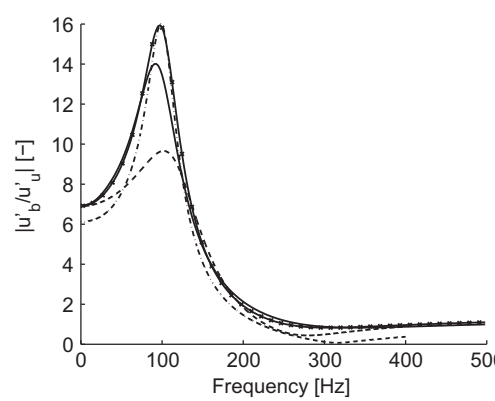

Fig. 6. Absolute value of transfer function for the velocity fluctuations. Solid line: numerical model with one-step chemistry. Solid line with symbols: numerical model with skeletal chemistry. Dashed line: analytical model. Dashed dotted line: McIntosh.

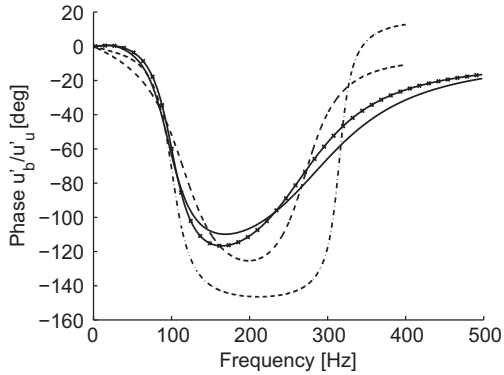

Fig. 7. Phase of transfer function for the velocity fluctuations. Solid line: numerical model with one-step chemistry. Solid line with symbols: numerical model with skeletal chemistry. Dashed line: analytical model. Dashed dotted line: McIntosh.

to the surrounding and on radiative properties of the burner, but is insensitive to fluctuating heat transfer due to the high specific heat of the material. Now, the acoustic response is described by T_{22} in which T_{surf} is not equal to T_{u} in the above equations:

$$T_{22} = \frac{T_{\mathrm{surf}}}{T_{\mathrm{u}}} + \frac{\overline{T}_{\mathrm{b}} - T_{\mathrm{surf}}}{T_{\mathrm{u}}}\mathscr{A}(\hat{\omega}) + \frac{T_{\mathrm{ad}} - \overline{T}_{\mathrm{u}}}{T_{\mathrm{u}}} \times \frac{1 - \mathscr{A}(\hat{\omega})}{\mathrm{i}\hat{\omega}}\frac{1}{2}\left(1 + \sqrt{1 + 4\mathrm{i}\hat{\omega}}\right), \tag{2.23}$$

where the stand-off distance is now given by

$$\frac{\psi_{\mathrm{f}}}{\delta_{\mathrm{T}}} = \ln(T_{\mathrm{ad}} - T_{\mathrm{u}})/(T_{\mathrm{ad}} - \overline{T}_{\mathrm{b}} + T_{\mathrm{surf}} - T_{\mathrm{u}}), \tag{2.24}$$

while T_{surf} is determined from the steady energy balance of the flame–burner system. This model predicts observations in experiments and numerical results much better, for instance the shift of the resonance to higher frequencies as T_{surf} increases due to decreasing stand-off distance can now be predicted.

2.5. Body forces for flame control

Another option to control burner surface stabilized flames externally is by using body forces. Two types of control using body forces are briefly reviewed. The first method is by using the influence of the gravitation and the second method is by applying an electrical field. The gravitation is always present in combustion devices and cannot be altered, but the flame stabilization and stability can be influenced by changing the orientation of the flame relative to the gravitational field (Landau–Darrieus instabilities, see next section). However, combustion in a rotating system is an example where the centrifugal force acts in a similar way and can be considered as a flame control mechanism. Differential body forces acting on different species act as drift velocities, which influence diffusion velocities. Such additional differential velocities may lead to separation of gases in high rotational force fields and to changes in the adiabatic burning velocity. This phenomenon has been studied recently in our group to support the development of a rotating burner for a micro-gas turbine, developed for heat and power generation.

The influence of a DC electric field on the combustion of planar burner surface stabilized flames has been studied mostly using burners with porous or perforated burner decks. In case of free flames, such as flames propagating in tubes, in combustion chambers or stabilized on Bunsen type burners, the influence of an electric field is mainly governed by field-induced body forces, known as ionic wind, acting on the flame as a whole without changing the internal structure noticeable. The complete structure is displaced due to the imposed momentum acting on the ions. One can expect a completely different situation in case of planar flames stabilized on burners, since the ionic wind would not affect the flame shape and area when the electric field is parallel to the direction of flame propagation. A question of interest is whether the accelerated electrons and ions are able to change the chemistry in the flame in such a way that the laminar burning velocity is changed. Early studies to investigate this [37,16,11] used the experimental technique developed by Botha and Spalding [14]. Measurements have shown that the change in s_{L} is minor and was at the limit of the accuracy of the measurement technique. Recently, the more accurate heat-flux method [39,72] has been used to investigate the change in burning velocity [12] and flame structure [128] of methane–air flames. The grounded burner plate served as the lower electrode and the upper electrode was either a non-cooled platinum wire screen or air cooled electrode made from a parallel row of thin tubes. Measurements showed a (limited) increase in the laminar burning velocity induced by the electric field which can reach values up to 8% for a stoichiometric mixture. An exponential relation between the applied voltage and the change in the adiabatic burning velocity s_{L} was found. Joule heating is not sufficient to explain these changes. Application of the electric field leads to a shift of the flame front towards the burner and to an increase of the burner deck temperature for constant gas velocity. Emission lines of excited nitrogen molecules were observed, but OH LIF measurements could not provide any significant indication of the influence on flame chemistry [128]. Accordingly, the available experimental

results are still insufficient to formulate unambiguous conclusions about the mechanisms involved.

The fact that a DC electric field affects the mean properties of planar stabilized flames can be used to control the dynamical behavior of these flames, for instance to influence the thermo-acoustic stability of combustion systems. Measurements, conducted for a range of flow rates and equivalence ratios in methane–air flames have shown that the application of a DC electric field leads to noticeable changes in the flame TF [129]. Particularly, the TF shifts in frequency domain to higher values when an electric field is applied. This shift, which can reach up to 40 Hz, relates to a decreased flame stand-off distance. The use of active control methods such as presented here, is still limited in combustion science. Their vast possibilities seem to be hardly realized now but are expected to become more important in sophisticated future combustion systems.

3. Multi-dimensional flames

This section deals with the influence of multi-dimensional effects on the propagation and dynamics of premixed laminar flames. On the smaller scales of the flame front, the multi-dimensional flame front can be curved and the approaching flow not homogeneous, for instance leading to local stretching or compression of the flame front. By introducing a local flame adapted coordinate system, the system can still be described by a quasi-one-dimensional system of equations with additional terms, related to flame stretching and curvature, reflecting transport phenomena along the flame front. This results in changes in the local burning velocity s_L, depending on the local thermo-diffusive nature of the flame front. On the larger scales, flames might display complicated dynamics, invoked by instabilities from a hydrodynamic nature, damped or enhanced by phenomena with a thermo-diffusive character. This might lead for instance to cellular structures and complicated dynamics of flames in external force fields, like acoustic waves. This subject is of importance for practical applications using laminar premixed flames, like Bunsen-type burners and combustors, but also for scientific reasons, because this knowledge is also the starting point of theories for the dynamics of turbulent flames.

These are the topics of study of Section 3. In the first Section 3.1, flame stretch and curvature phenomena are reviewed and the quasi-one-dimensional set of so-called flamelet equations is presented. The analysis of this set of equations is presented in Section 3.2 by reviewing the effects of these distortions on the burning velocity. This behavior is subsequently used in Section 3.3 to study flame stability and to display their effect on flame dynamics. State-of the art knowledge of flame dynamics in an acoustic field is reviewed in Section 3.4 while other physical control mechanisms, based on external (electrical and gravitational) fields will be briefly considered in Section 3.5.

3.1. Flame stretch

It is well known that flame stretch, caused for instance by local velocity gradients near the flame front, may have an important influence on premixed laminar flames. This was first recognized by Karlovitz and Denniston [55] and Markstein [76]. Flames, experiencing flame stretch have been studied by a lot of researchers since, such as Buckmaster, Matalon, Matkowsky, Joulin, Clavin, Chung and Law [17,18,51,77,78,24]. A rigorous mathematical analysis of stretched flames, based on matched asymptotic expansions, was given by Matalon and Matkowsky [77] and Joulin and Clavin [51] simultaneously . An integral analysis of flame stretch was presented by Chung and Law [24]. Flame sheet models with one-step chemistry and a single Lewis number are frequently used to study stretched flames. Stimulated by the ideas of Buckmaster [17], who first realized that variations in the preheating zone thickness are of importance and using ideas of the integral analysis of Chung and Law [24], de Goey et al. [40,41] applied the flame stretch formalism to flames with finite flame front thickness.

We base this review on the more fundamental formulation of de Goey et al. [40] and relations with earlier work will be explained. The governing equations are formulated in a flame adapted coordinate system, from which a set of quasi-one-dimensional flamelet equations arises with extra source terms, expressed in terms of the flame stretch rate. On the basis of the flame adapted coordinate system, the full system of equations can be split in three parts as follows: (a) the mass and momentum equations describing fluid flow, (b) the kinematic condition, formulated as a generalized G-equation [133], describing the flame motion, and (c) the flamelet system for the inner flame structure and burning velocity.

The flame adapted coordinate system is defined in the following way. The flame zone is defined in terms of some progress variable $\mathscr{Y}$, strictly increasing between the unburned ($\mathscr{Y}_u$) and burned ($\mathscr{Y}_b$) flame boundaries (see Fig. 8). For $\mathscr{Y}$ we may take the temperature or any combination of species mass fractions, provided $\nabla\mathscr{Y} \neq \mathbf{0}$ everywhere in the flame. Next, we introduce the curvilinear coordinate system $\boldsymbol{\xi} = (\xi^1, \xi^2, \xi^3)$, referred to as *flame coordinates*, by the relations

$$\mathscr{Y}(x_1, x_2, x_3, t) = \mathscr{Y}(\xi^1), \quad \frac{\partial \boldsymbol{x}}{\partial \xi^1} \cdot \frac{\partial \boldsymbol{x}}{\partial \xi^\alpha} = 0 \quad (\alpha = 2, 3) \quad \text{for} \quad \mathscr{Y}_u \leqslant \mathscr{Y} \leqslant \mathscr{Y}_b, \quad (3.1)$$

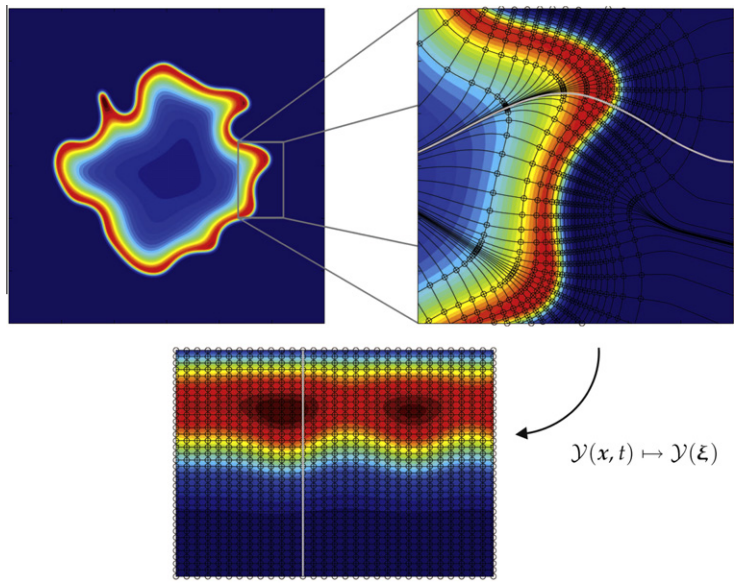

Fig. 8. Visualization of the temperature field in a premixed flame (left top) with a zoom including the flame contours (constant $\mathscr{Y}$-values) and flamelet paths (perpendicular to flame contours) in a Cartesian coordinate system (right top) and temperature field in flame adapted system (bottom).

i.e., coordinate surfaces $\xi^1 = \text{Const}$ coincide with flame surfaces $\mathscr{Y}(\boldsymbol{x},t) = \text{Const}$ and the ξ^1-coordinate lines are orthogonal to these. The flame surfaces move with velocity $\boldsymbol{v}_\text{f} = \boldsymbol{v} + s_\text{L}\boldsymbol{n}$, where $\boldsymbol{n}$ is the unit normal (on the flame surfaces) directed towards the unburned gas mixture, resulting in a time-dependent coordinate system $\boldsymbol{\xi} = \boldsymbol{\xi}(\boldsymbol{x},t)$. The motion of the flame surfaces satisfies the kinematic condition

$$\frac{\partial \mathscr{Y}}{\partial t} + \boldsymbol{v}\cdot\nabla\mathscr{Y} = s_\text{L}|\nabla\mathscr{Y}|, \tag{3.2}$$

which is reminiscent of the G-equation [133]. This equation holds in the *entire* flame zone and not just on an infinitely thin flame sheet. The consequence however is that each iso-plane moves with its own velocity, meaning that $\boldsymbol{v}$ and s_L are local quantities and are different for each iso-plane of $\mathscr{Y}$.

Due to flow straining and flame curvature, the local flame structure and burning velocity will change. These effects are included in the stretch rate K_σ [66,78], defined for a flame surface as the fractional rate of change of area of an infinitesimal surface element $\mathrm{d}S = \sigma\,\mathrm{d}\xi^2\,\mathrm{d}\xi^3$, i.e., $K_\sigma = 1/\sigma\,\mathrm{d}\sigma/\mathrm{d}t$. Traditionally, K_σ is defined on the flame front, regarded as an interface separating unburnt and burnt gases. However, flow and flame distortions might also compress or expand the flame front in ξ^1-direction, i.e., it might change the thickness of the flame (see also Sections 2.3 and 2.4). As an extension for flames of finite thickness, the mass-based stretch rate K [40,41,123] is introduced, which is defined as the fractional rate of change of mass contained in an infinitesimal volume element $\mathrm{d}V = \sqrt{g}\,\mathrm{d}\xi^1\,\mathrm{d}\xi^2\,\mathrm{d}\xi^3$ moving with velocity $\boldsymbol{v}_\text{f}$. Formulated in flame coordinates, the expression for K reads

$$\begin{aligned} K &= \frac{1}{M}\frac{\mathrm{d}M}{\mathrm{d}t} = \frac{1}{\sqrt{g}}\frac{\mathrm{d}}{\mathrm{d}t}\left(\sqrt{g}\right) + \frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t} \\ &= K_\sigma + \frac{1}{h_1}\frac{\mathrm{d}h_1}{\mathrm{d}t} + \frac{1}{\rho}\frac{\mathrm{d}\rho}{\mathrm{d}t}. \end{aligned} \tag{3.3}$$

Thus, compared to K_σ, K includes additional terms, e.g., due to flame thickness variations (with scale factor $h_1 = \left|\frac{\partial \boldsymbol{x}}{\partial \xi^1}\right| = \sqrt{g}/\sigma$ in ξ^1-direction). It is important to note that K, like $\boldsymbol{v}$ and s_L, is a scalar field defined in the *entire* flame zone and not just on an infinitely thin flame surface like K_σ.

Combining the definition of K with the mass continuity equation and using the relation $\boldsymbol{v}_\text{f} = \boldsymbol{v} + s_\text{L}\boldsymbol{n}$ we get

$$\nabla\cdot(m\boldsymbol{n}) = -\rho K, \tag{3.4}$$

where we introduced $m = \rho s_\text{L}$ as the local 'mass burning rate' of the flame iso-plane $\mathscr{Y}$. This equation states that K is responsible for changes in m along the flamelet through the flame. In flame coordinates, we obtain the following set of conservation equations for mass, species mass fractions and energy (temperature):

$$\frac{\partial}{\partial s}(\sigma m) = -\sigma\rho K, \tag{3.5a}$$

$$\begin{aligned} \frac{\partial}{\partial s}(\sigma m\mathscr{Y}) - \frac{\partial}{\partial s}\left(\sigma\rho D_\mathscr{Y}\frac{\partial\mathscr{Y}}{\partial s}\right) \\ = \sigma(\omega_\mathscr{Y} - \rho K c_p\mathscr{Y}), \end{aligned} \tag{3.5b}$$

$$\frac{\partial}{\partial s}(\sigma m Y_i) - \frac{1}{\mathrm{Le}_i}\frac{\partial}{\partial s}\left(\sigma\frac{\lambda}{c_p}\frac{\partial Y_i}{\partial s}\right) = \sigma(\omega_i - \rho K Y_i), \quad i = 1,\ldots,N_\mathrm{s}, \tag{3.5c}$$

$$\frac{\partial}{\partial s}(\sigma m T) - \frac{\partial}{\partial s}\left(\sigma\frac{\lambda}{c_p}\frac{\partial T}{\partial s}\right) = \sigma(\omega_\mathrm{T} - \rho K T), \tag{3.5d}$$

where $\mathrm{d}s = h_1\,\mathrm{d}\xi^1$ is the arclength along the ξ^1-coordinate lines. In the derivation of system (3.5) it is assumed that diffusive fluxes tangent to the flame surfaces are negligible compared to the stretch terms proportional to K, which is justified in the flamelet regime [40]. We can derive similar equations for the (specific) enthalpy h and the element mass fractions Z_j. The solution of this so-called flamelet system (3.5d) includes the flame structure in terms of $Y_i(s)$, $T(s)$ and the mass burning rate $m(s) = \rho s_\mathrm{L}$ of the flamelet.

3.2. Effects of stretch and curvature on flame propagation

In the present subsection, we analyze the solution of the flamelet system, where it is assumed that the stretch rate K and the curvature σ are given functions of s. This functional dependence, in fact, is determined by the local conditions of the flamelet in a given flow field. Note that $K(s)$ is close to a constant for weakly strained and curved flames in a flow field with large scale distortions. For strong stretch and curvature, $K(s)$ and $\sigma(s)$ can be strongly varying functions of s. We will derive general expressions for the change in mass burning rate and also consider some special cases. The focus is on quantitative physical descriptions of the phenomena.

Analogous to the derivation of (2.7), multiplying the flamelet equations (3.5a) with $\sigma\partial\mathscr{Y}/\partial s$ and integrating across the flame, both for $K = 0$ and $K \neq 0$, we obtain the relation

$$\left(1 + \frac{\mathrm{Ka}_{\mathscr{Y}}}{\mathrm{Le}_{\mathscr{Y}}}\right) m_\mathrm{b}(\boldsymbol{\psi}_\mathrm{b}) = m_\mathrm{b}^0(\boldsymbol{\psi}_\mathrm{b}), \tag{3.6}$$

for the mass burning rate m_b at the burned side of the flame and its stretchless counterpart m_b^0, where $\mathrm{Ka}_{\mathscr{Y}}$ is the Karlovitz integral for $\mathscr{Y}$, given by

$$\mathrm{Ka}_{\mathscr{Y}} = \frac{\mathrm{Le}_{\mathscr{Y}}}{(\sigma m)_\mathrm{b}}\int_{s_\mathrm{u}}^{s_\mathrm{b}} \sigma\rho K\widetilde{\mathscr{Y}}\,\mathrm{d}s, \quad \widetilde{\mathscr{Y}}(s): = \frac{\mathscr{Y}(s) - \mathscr{Y}_\mathrm{u}}{\mathscr{Y}_\mathrm{b} - \mathscr{Y}_\mathrm{u}}. \tag{3.7}$$

The mass burning rates are assumed to be functions of the state variables $\boldsymbol{\psi}_\mathrm{b} = (h_\mathrm{b}, Z_{j,\mathrm{b}})$, describing the state of the burnt equilibrium mixture [42]. Note that $\boldsymbol{\psi}_\mathrm{b}$ as argument of m_b^0 corresponds to a *stretched* flame and that $m_\mathrm{b}^0 = m_\mathrm{b}^0(\boldsymbol{\psi}_\mathrm{b}^0)$. For weakly stretched flames we can expand $m_\mathrm{b}^0(\boldsymbol{\psi}_\mathrm{b})$ in a first order Taylor series around the stretchless state $\boldsymbol{\psi}_\mathrm{b}^0$, to obtain

$$\frac{m_\mathrm{b} - m_\mathrm{b}^0}{m_\mathrm{b}^0} = -\frac{\mathrm{Ka}_{\mathscr{Y}}}{\mathrm{Le}_{\mathscr{Y}}} + \Delta h_\mathrm{b}\frac{\partial}{\partial h_\mathrm{b}}(\ln m_\mathrm{b}^0) + \sum_{j=1}^{N_\mathrm{e}}\Delta Z_{j,\mathrm{b}}\frac{\partial}{\partial Z_j}(\ln m_\mathrm{b}^0) + \text{hot}, \tag{3.8}$$

where $\Delta h_\mathrm{b} = h_\mathrm{b} - h_\mathrm{b}^0$ and $\Delta Z_{j,\mathrm{b}} = Z_{j,\mathrm{b}} - Z_{j,\mathrm{b}}^0$. In order to determine the increments Δh_b and $\Delta Z_{j,\mathrm{b}}$, we have to integrate the flamelet equations for h and Z_j across the flame. Assuming that all diffusive fluxes vanish at the unburned and burned sides of the flame, we find [40]

$$\left(1 + \frac{\mathrm{Ka}_{\mathscr{Y}}}{\mathrm{Le}_{\mathscr{Y}}}\right)\Delta h_\mathrm{b} = -\sum_{i=1}^{N_\mathrm{s}}\left(\frac{\mathrm{Ka}_i}{\mathrm{Le}_i} - \frac{\mathrm{Ka}_{\mathscr{Y}}}{\mathrm{Le}_{\mathscr{Y}}}\right)h_i^0(Y_{i,\mathrm{b}} - Y_{i,\mathrm{u}}), \tag{3.9}$$

where Ka_i is the Karlovitz integral of the ith species, defined analogous to $\mathrm{Ka}_{\mathscr{Y}}$ in (3.7), with $\widetilde{\mathscr{Y}}(s)$ replaced by $\mathrm{Le}_i\widetilde{Y}_i(s)$. Similar expressions hold for $\Delta Z_{j,\mathrm{b}}$. Thus, changes in enthalpy and element mass fraction are due to a combination of flame stretch and preferential diffusion.

It is important to know at this stage, how accurate the expressions (3.6)–(3.9) are. In [96], Eq. (3.6) has been used to analyze turbulent flame kernels in the thin reaction zones regime. Even for very highly stretched and curved flamelets, Eq. (3.6) predicts the numerically computed changes in mass burning rate very accurately for the case that all Lewis numbers $\mathrm{Le}_i = 1$. This formulation might therefore be the starting point of a strong stretch theory. Results for the general case $\mathrm{Le}_i \neq 1$ are not yet thoroughly compared with detailed numerical results. As a first result, Eq. (3.8) in combination with relation (3.9) for Δh_b and similar expressions for $\Delta Z_{j,\mathrm{b}}$ have been validated with numerically computed mass burning rates of weakly strained planar flames with detailed kinetics and constant Lewis numbers [122]. To evaluate the Karlovitz integrals in the weak stretch limit we may replace ρ, $\widetilde{\mathscr{Y}}$ and m in (3.7) by their stretchless equivalents ρ^0, $\widetilde{\mathscr{Y}}^0$ and m^0, determined from the flamelet equations (3.5d) with $K = 0$. This comparison has shown that these equations are able to predict all important changes in element mass fractions, enthalpy and mass burning rate in numerically computed stretched flames. It seems that the Lewis numbers of all species take part in the preferential diffusion effects. These last equations therefore seem complicated and hard to apply to the general case of flames described by multiple species chemistry and transport.

We therefore first consider the simplified case of a stagnation flame, governed by a one-step irreversible reaction $\mathscr{F} \to \mathscr{P}$ with a single Lewis number Le and an infinitely thin reaction layer [123]. In this case we find that $\Delta Z_{j,\mathrm{b}} = 0$, leading to

$$\frac{m_b - m_b^0}{m_b^0} = -\mathscr{M}_b \mathrm{Ka}_b + \mathscr{O}(\mathrm{Ka}_b^2), \tag{3.10}$$

where the Karlovitz number is given by $\mathrm{Ka}_b = \rho_b K \delta_T / m_b^0$, $K = a$ being the (constant) stretch rate, and $\mathscr{M}_b$ is the Markstein number, given by

$$\mathscr{M}_b = (\theta+1)\left(\frac{1}{\theta}\log(\theta+1) + \frac{1}{2}\mathrm{Ze}(\mathrm{Le}-1) \times \frac{1}{\theta}\int_0^1 v^{\mathrm{Le}-2}\log(1+\theta v)\,\mathrm{d}v\right), \tag{3.11}$$

with $\theta = (T_b^0 - T_u)/T_u$ the thermal expansion coefficient. The first and second term in (3.11) describe the 'direct' stretch effect and the combined effect of preferential diffusion in combination with flame stretch [42].

An alternative approach to derive (3.10) and (3.11) is presented by Clavin [28]. He considers the flame as a surface separating the (incompressible) burned and unburned gas mixtures and the combustion equations are integrated across the reaction layer to give boundary and jump conditions for the governing equations in the (un)burned mixtures, which are subsequently solved using matched asymptotic expansions ($\mathrm{Ze} \to \infty$). He obtained an equivalent result, except that in the integral in Eq. (3.11) Le has to be replaced by 1. However, in his derivation it was assumed that $\mathrm{Le} - 1 = \mathscr{O}(1/\mathrm{Ze})$, so that this result is in lowest order in $1/\mathrm{Ze}$ equal to ours. A similar expression was derived by Matalon and Matkowski [77]. Rogg and Peters performed a weak stretch analysis for a stoichiometric methane–air flame described by a systematically reduced 4-step kinetic model and found a similar expression as Eq. (3.11). The sensitivity of the mass burning rate to the temperature, i.e., the coefficient $\mathrm{Ze}/2 = (T_{ad} - T_u)\partial \ln m_b / \partial T_{ad}$ in Eq. (3.11), is replaced by $2(T_{ad} - T_u)/(T_{ad} - T_0)$, similar to Eq. (2.6).

An interesting alternative is based on the intrinsic disturbed flame equations (IDFE) by Class et al. [26,27], which is a quasi-one-dimensional system of conservation laws, analogous to the flamelet equations (3.5d), although formulated in terms of different flame coordinates. The flame itself is represented by the surface $\xi^1 = 0$ separating the burned and unburned gas mixtures. They apply IA to the IDFE, both with and without reaction terms, to derive jump conditions that replace (the effect of) the reaction terms. Subsequently, solving the IDFE coupled with the jump conditions by matched asymptotic expansion gives a relation for the laminar burning speed s_L, which has to be used in (3.2) to determine the propagation of the flame.

The question remains how well the expressions (3.6)–(3.9) describe stretch related phenomena in flames governed by realistic chemistry and multi-species transport. In case of constant Lewis numbers unequal to 1 for all species, Groot et al. [43] have shown that the terms describing the effect of changes in enthalpy and elements due to differential diffusion, can be rewritten as a combination of terms related to changes in flame temperature and major species products H_2O, CO_2 and O_2. Furthermore, recent studies indicate that the sensitivity of the burning velocity to changes in flame temperature is dominant. These temperature variations are induced by a combination of changes in enthalpy and element mass fractions (mainly equivalence ratio). For fuel lean flames, Groot et al. have shown that all Lewis numbers except the fuel Lewis number cancel. This then remarkably reduces to the single unit Lewis number expression (3.11) for $\mathscr{M}_b$, where the Lewis number equals the fuel Lewis number $\mathrm{Le}_{\mathrm{Fuel}}$. For stoichiometric and rich mixtures, similar expressions are not valid. The reason for this is that both fuel and oxidizer mass fractions vanish in the reaction layer for stoichiometric and rich flames, so that both Lewis numbers have to play a role in its diffusive transport towards the inner layer. Furthermore, CO and H_2 are produced in such flames, so that Lewis numbers of these intermediates are also expected to be present. Which of these effects dominates is not clear yet. This explains why one-step expressions using the oxidizer Lewis number or a combination of fuel and oxidizer Lewis numbers cannot reproduce numerical results for stoichiometric or rich flames as proposed in one-step chemistry models [52]. Note that Groot et al. [43] have also shown that Markstein numbers of pure curvature and strain parts in flames might be different, but the combined influence is described by the above models.

In order to get more insight in the different contributions of non-unit Lewis number effects on the propagation velocity of stretched flames, as an example we present a weak stretch analysis of the mass burning rate of lean ($\Phi = 0.6$) and rich ($\Phi = 1.4$) propane–air flames, computed numerically with a detailed reaction mechanism [114]. In the simulations, one-dimensional weakly-stretched steady flames ($K(s) = a$) are computed with constant Lewis numbers for all species. The numerical mass burning rate m_b is shown as function of the Karlovitz number Ka_b in Fig. 9 with black markers. To understand the impact of different assumptions in the Markstein number expressions, several assumptions in the derivation of (3.11) are relaxed. To start with, we consider the expressions (3.8) and (3.9) valid for weakly stretched flames, where the burned side of the flame at $s = s_b$ is chosen at the location where the fuel consumption rate has decreased to 5% of its maximum value in the flame. For the progress variable we choose $\mathscr{Y} = Y_{C_3H_8}$. This result is shown in the figure with a thick solid curves. Note that this expression quite accurately reproduces numerical results. On the other hand,

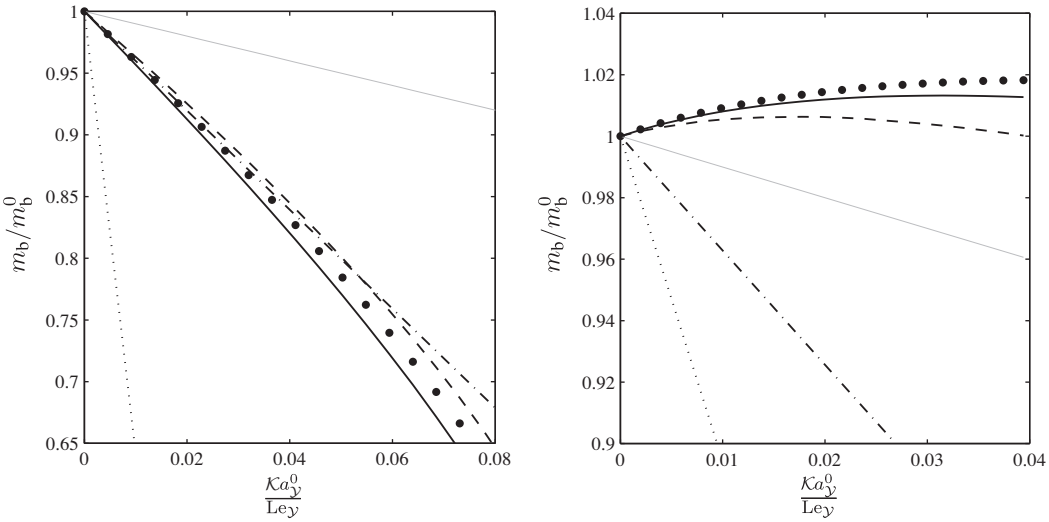

Fig. 9. Mass burning rate of stretched propane–air flames. Left: Lean mixture ($\Phi = 0.6$). Right: Rich mixture ($\Phi = 1.4$). Spherical markers: numerical simulations. Thick drawn curves: Eq. (3.8). Thin drawn curves: $m_b/m_b^0 = 1 - \mathrm{Ka}_{\mathcal{Y}}^0/\mathrm{Le}_{\mathcal{Y}}$. Dashed curves: Eq. (3.12). Dash-dotted curves: Eq. (3.13). Dotted curves: Eq. (3.11).

thin solid lines represent the unit Lewis number result of Eq. (3.8) without preferential diffusion, i.e., $\Delta h_b = \Delta Z_{j,b} = 0$. The numerical result is not at all reproduced.

The enthalpy and element mass fraction changes can be written in terms of changes in temperature and mass fractions of the major species H_2O, CO_2 and O_2 as mentioned. Neglecting the species' terms as indicated above and taking only variations in T_b into account gives:

$$\frac{m_b}{m_b^0} = 1 - \frac{\mathrm{Ka}_{\mathcal{Y}}^0}{\mathrm{Le}_{\mathcal{Y}}} + \frac{\mathrm{Ze}}{2}\frac{\Delta T_b}{T_{ad}^0 - T_u}. \tag{3.12}$$

Taking ΔT_b from the numerical solution of (3.5d) gives the dashed curves in the figure. This approximation is accurate for both the lean and rich case, indeed. Next, assuming one-step chemistry, the temperature change can be expressed as the difference between the Karlovitz integrals of temperature and the deficient (lean) species:

$$\frac{m_b}{m_b^0} = 1 - \frac{\mathrm{Ka}_{\mathcal{Y}}^0}{\mathrm{Le}_{\mathcal{Y}}} + \frac{\mathrm{Ze}}{2}\left(\frac{\mathrm{Ka}_{\mathrm{lean}}^0}{\mathrm{Le}_{\mathrm{lean}}} - \mathrm{Ka}_T^0\right), \tag{3.13}$$

where 'lean' refers to the Karlovitz integral of the lean species (either fuel or oxidizer). This relation is plotted with a dash-dotted curves. It works well for the lean case but not for the rich (as expected). Finally, assuming analytical (exponential) profiles for ρ^0, T^0 and Y_i^0 finally gives Eq. (3.11). This expression is shown with dotted lines and is a very poor representation of the true result. Note that a much more accurate evaluation can be obtained substituting the one-dimensional numerical solution of (3.5d) for ρ, T and Y_i instead of exponential solutions.

Burning velocities and Markstein numbers have also been determined experimentally for many mixtures. For instance, during experiments on laminar burning velocity using the opposed-jet flames and constant volume method, corrections for flame stretch are needed, which enables the evaluation of Markstein numbers at the same time [44]. Linear correction for stretch effects are mostly used while non-linear correction is mostly more accurate. Reliability is therefore sometimes rather limited both for laminar burning velocities as well as the measured Markstein numbers. In future, accuracy should be improved, e.g., by using the heat-flux method, where an accuracy in the burning velocity of the order of 1% is within reach. For Markstein numbers, current accuracy is unclear, because reported values depend largely on the position within the flame front where they are evaluated [124]. This position is often not accurately known and/or mostly not reported. Experimentalists are urged to report the corresponding position for making a fair comparison with models and other experimental data possible. More reliable experimental data and a more thorough confrontation with numerical results and analytical expressions are then possible. It is recommended to update chemical kinetic mechanisms and transport models from simultaneously obtained accurate burning velocity and Markstein number data in future studies. Flame stretch in curved flames are not yet thoroughly analyzed although Eqs. (3.6) and (3.8) are valid for curved

flames as well. Extension of the analysis to curved flames is therefore recommended as well.

3.3. Thermal-diffusive effects and flame stability

The previous subsection focussed on the small-scale physical and chemical processes inside the flame front. In the present subsection we will consider what happens with the flame on a larger scale. It is well known that, depending on the Markstein number, flames may show different kinds of intrinsic instability. The formation of cellular flames was shown experimentally in a very convincing way by Clanet and Searby [25]. Flame front instabilities have been studied in many papers and reviews are given in, e.g., [28,68,79,118].

The starting point of the analysis is the well-known theoretical studies of Darrieus [30] and Landau [65], who independently predicted that planar flames are unconditionally unstable due to thermal expansion of the gas, also known as the hydrodynamic instability. The Landau–Darrieus (LD) theory assumes an infinitely thin flame front with a constant burning velocity s_L and thermo-diffusive processes inside the front are discarded. If the planar flame is perturbed with alternating convex and concave segments towards the unburnt mixture, then streamlines are bend towards the flame normal due to thermal expansion. Therefore, the streamlines must diverge in front of a convex region, resulting in a decreasing local velocity. As a result the burning velocity will be larger than the gas velocity and the perturbation will grow. Mathematically, the instability is found by expanding the perturbation in normal modes of the form $\exp(\omega t + \mathrm{i}k\xi)$, with ω the growth rate, ξ the transverse coordinate and k the wave number. Substitution in the conservation equations leads to a dispersion relation $\omega = \omega(k)$. Without gravity, the linear dispersion relation reads

$$\omega(k) = ks_L\omega_{LD}, \quad \omega_{LD} = \frac{\tau}{1+\tau}\left(\sqrt{1+\tau-1/\tau}-1\right), \tag{3.14}$$

with $\tau = \rho_u/\rho_b$ the density ratio. Note that the growth rate ω is positive for all wave numbers since $\tau > 1$. If gravitation is included, this leads to an additional stabilizing part in ω_{LD} for long wavelengths, depending on the orientation. At smaller wavelengths the hydrodynamic instability will always prevail.

In practice, however, stable planar flames do exist. The reason for this discrepancy is related to the thermo-diffusive processes inside the flame front, characterized by the influence of flame stretch on the burning velocity as described in the previous subsection. When thermal-diffusive effects inside the flame front are taken into account, Eq. (3.14) is extended by additional terms:

$$\omega(k) = ks_L\omega_{LD} - k^2 s_L \delta_T\left(B_1 + B_2\frac{1}{2}\mathrm{Ze}(\mathrm{Le}-1)\right), \tag{3.15}$$

with $B_i > 0$ $(i = 1, 2)$ constants [79]. The term with B_1 is related to the direct stretch effect and the B_2-term is related to the preferential diffusion effect as discussed in the previous subsection. First, let us consider equi-diffusive flames ($\mathrm{Le} = 1$), for which case the burning velocity always decreases with increasing stretch rate. Since a convex region of the perturbed planar flame is positively stretched by flow straining and flame curvature, its burning velocity decreases, which has a stabilizing effect as seen in Eq. (3.15). For non-equi-diffusive flames there is an additional effect due to the combination of stretch and the imbalance of diffusion of mass and heat. For an effective Lewis number $\mathrm{Le} > 1$, positive stretch induces a lower flame temperature and an even lower burning velocity; see previous subsection. Thus, at a convex region, where $K > 0$, the gas velocity decreases due to the hydrodynamic instability, but the burning velocity decreases as well due to direct stretch and preferential diffusion effects. These stretch related effects damp the hydrodynamic instability if the term with δ_T is positive, or in other words, if the effective Lewis number exceeds a critical value Le_c. On the other hand, for $\mathrm{Le} < \mathrm{Le}_c$ the Markstein number is negative and the short wavelength perturbations are also unstable and the hydrodynamic instability is enhanced by the thermal-diffusive effects. Since this so-called thermal-diffusive instability is caused by a change of the diffusive structure of the flame, the cell size of the instability is expected to be one order larger than the flame thickness. An additional δ_T-term can be introduced in Eq. (3.15) to describe viscous effects inside the flame front [79]. It should be noted finally, that heat losses increase the unstable combustion regime, like in the case of planar flame pulsations.

From the previous one can conclude that the effective Lewis number, or equivalently, effective Markstein number as related by Eq. (3.11), is a key parameter of the thermal-diffusive instability. For a one-step reaction between fuel and oxidizer, this effective Lewis number is found to be a weighted average of the individual Lewis numbers Le_F and Le_{O_2} [52]. It changes with equivalence ratio from Le_F for fuel-lean mixtures to Le_{O_2} for fuel-rich mixtures, indicating that the diffusivity of the deficient reactant determines the characteristics of thermal-diffusive effects. This seems to agree with effective Lewis numbers for lean hydrogen and propane mixtures extracted from simulations of symmetrical counterflow flames [121]. For lean multi-component fuel mixtures, Law et al. [69] have shown that the effective Lewis number

reduces to the weighted average of the Lewis numbers of the different fuel components, the weights determined by their contribution to the total heat release. In the previous subsection, it has been shown that this model is not valid to predict Markstein numbers in stoichiometric and rich flames described by multiple species and multiple Lewis numbers. For the propane-air flames considered in the previous subsection, Markstein numbers are positive and large for lean mixtures but negative and large (in absolute value) for rich mixtures. The lean flame is therefore predicted to be stable, while the rich flame is instable.

Matalon [79] recently reviewed current models available for predicting premixed flame stability. Dispersion relations like Eq. (3.15) are only available for some special cases like a perturbed planar flame and a spherically propagating flame. For more complex flame shapes, less is known. Furthermore, these expressions explicitly rely on small distortions, while it is known that nonlinear effects are very important to describe flame dynamics. For more general cases, non-linear versions of the dispersion relations should be used. An alternative is the use of numerical models with reliable submodels for the influence of small-scale stretch effects on the mass burning rate as derived in the previous subsection. This especially holds for flames in turbulent flows. A very popular model is the G-equation model with a realistic description for the influence of the stretch and curvature on flame dynamics [99]. Note that such models are only valid if all flow scales are much larger than the flame thickness. In case of small-scale perturbations, inhomogeneous stretch and curvature fields perturb the flame structure and the model breaks down, including the weak stretch theory. These small-scale perturbations can be taken into account, but then one has to rely on models where the flame front thickness δ_T is resolved, like so-called laminar flamelet models.

3.4. Flame dynamics in acoustic fields

An interesting and thoroughly studied example where the intricate interaction between the premixed flame front and external perturbations is of importance, is the dynamics of Bunsen-type flames in an external acoustic field. An example of a Bunsen flame, excited with acoustic waves with different perturbation frequencies is presented in Fig. 10. This subject is also of interest from a practical point of view as mentioned in Section 2.4. The premixed Bunsen-type flame is a traditional and well-studied object in combustion science. Furthermore, the relative simplicity of this flame allows a detailed understanding of the interaction with an acoustic field, which is beneficial for the comprehension of more complicated flames. Accordingly, both fundamental and practical interest has stimulated the research of forced dynamics of such flames. The modeling of Bunsen flame dynamics in an acoustic field is considered here after a review of previous, mostly experimental, work. As in Section 2.4, the transfer matrix equation (2.17) and flame TF equation (2.18) describe the transfer of acoustic waves, since the flames are generally small in size compared to the acoustic wave lengths involved.

The systematic experimental study of many different combustion systems equipped with burners of multiple laminar Bunsen type flames goes back to the 1950s; see [101] for an overview. Later, Matsui and Sugimoto [82,120] studied acoustically excited flames situated on a range of burner patterns, and more recently, the acoustic TF of multiple flames anchored on a perforated burner deck was studied by Noiray and Durox [91,92]. From this work it has been established that the TF gain shows a low-pass filtering behavior: it tends to 0 for increasing frequency in a non-monotonic way with well recognizable minima and maxima. The phase lag grows linearly as a function of frequency, which is typical for a system with time delay. Comparing the TF phase time lag with the convective time delay $\tau_c = h/\bar{u}$, where h is the flame height and $\bar{u}$ the average flow velocity, a good correlation between τ_c and the TF time delay was found.

The experimental investigation of the TF of a single Bunsen-type flame has a shorter history than the study of multiple flames. Results of direct

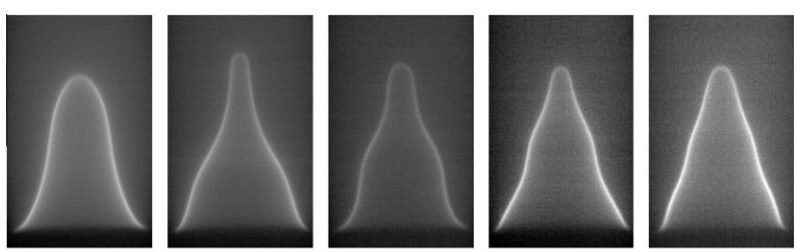

Fig. 10. Instant images of an excited Bunsen flame cone for different perturbation frequencies. The flame is stabilized on the burner with a diameter of 1 cm with a flat velocity profile, $\bar{u} = 100$ cm/s, $\Phi = 1.2$. Velocity excitation amplitude $\hat{u} \approx 20$ cm/s.

measurements of the TF were first reported by Ducruix and Schuller [32], while non-linearities were studied systematically by Durox et al. [31] and a systematic parametric study of the TF of single Bunsen flames was done recently [59]. Besides the already mentioned time delay property of the TF phase and the low-pass filtering behavior of the TF gain (see Fig. 11), the presence of a weakly frequency dependent component has been recognized in the TF (for instance, notice the saturation point in the polar plot Fig. 11c). This was evident from the high frequency saturation (set-off) followed by a very slow drift of the TF phase and gain. For single flames it has been shown that it is possible to generalize the TF behavior in the form of a universal empirical expression. The correlations found between the burner/flame parameters and the TF coefficients allows an *a priori* prediction of the TF time delay, low-pass filtering behavior and off-set. Another feature, typical for multiple flame configurations but absent for a single Bunsen-type flame, is the overshoot of the TF gain above 1 for some excitation frequencies. In spite of the qualitative resemblance of single and multiple flame TF's, there is a difference due to the interaction of the flames that leads to a change in flame shape and height. For instance, the flame foot motion, identified in [60] as one of the possible key elements responsible for the observed phase of the flame TF, is expected to be quite different. Accordingly, one more burner parameter, viz the inter-flame spacing (pitch) becomes important for the TF of a multiple flame burner [62,63].

Extension of the linear TF to a nonlinear description of the flame dynamics leads to the concept of an amplitude dependent TF, the so-called flame describing function (FDF) [31,56,93]. A procedure to measure the FDF is almost the same as for the linear TF, but the amplitude of the flow velocity perturbation is the additional parameter. The concept of FDF can be applied to determine, for example, the limit cycle amplitude, form, finite amplitude triggering of instability, hysteresis and mode switching of the instabilities, as illustrated in [93,95] and at the present symposium in [15].

Apart from the global characterization of the thermo-acoustic behavior in terms of the TF/FDF, a lot of research has been done on the analysis of detailed kinematics of Bunsen-type flames on top of the corresponding burners (see Fig. 10), because this forms the basis of the observed TF/FDF behavior. Probably the first

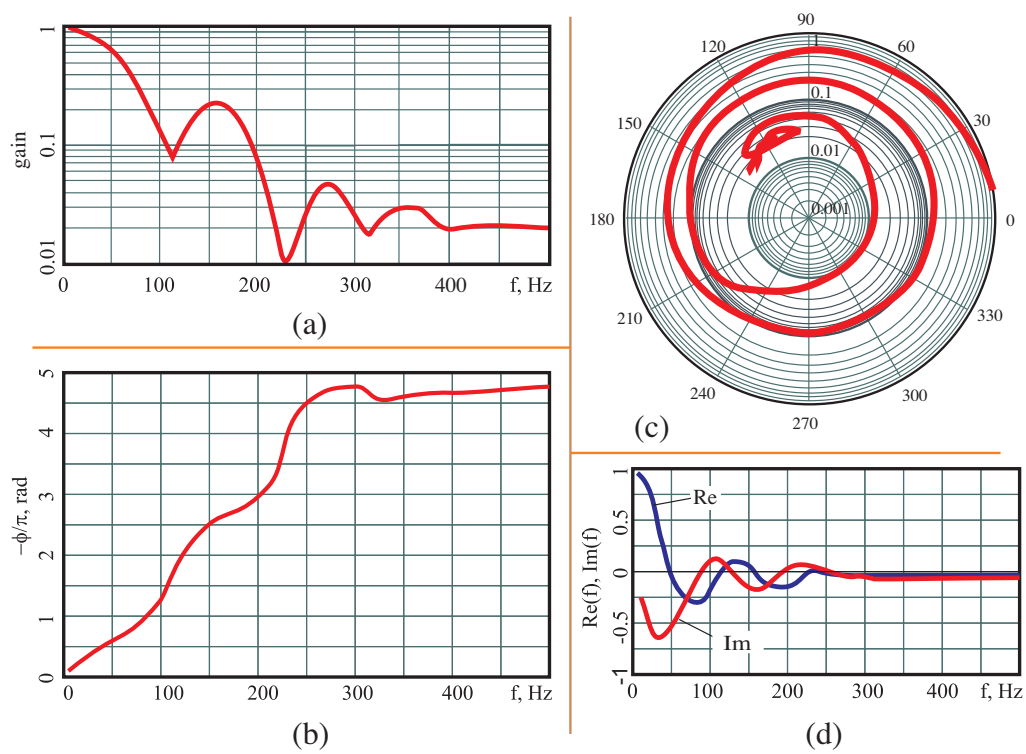

Fig. 11. Flame TF for a 1 cm diameter burner with a parabolic mean velocity profile, $\bar{u} = 150$ cm/s, $\Phi = 1$; (a) gain of the TF; (b) phase delay (absolute value); (c) the TF in polar plot (module of the phase delay corresponds to the polar angle); (d) Re and Im parts of the TF.

clear phase resolved pictures of an excited Bunsen-type flame were published by Markstein [75] and later by Blackshear [9]. Later, acoustically excited flame front kinematics was studied elaborately in the EM2C laboratory (see for instance [22,109,110]). It has been observed that the flame, subject to a weak flow perturbation, oscillates periodically and with the same frequency as the perturbation frequency around its mean position. Flame excitations appear at the flame foot and travel along the front to the tip with a velocity which is of the order of the convective flow velocity. The amount of flame wrinkles thus depends on the applied frequency. Results of further experiments [59,61,63] also indicate that the flame anchoring point moves (predominantly in the radial direction) and that the magnitude of the flame foot motion is frequency dependent. It was also found that the oscillation of the flame surface area correlates with the heat release rate which supports the assumption of a constant s_L along the front. From this it is clear that flame surface area fluctuations govern the heat-release fluctuations (instead of the heat loss fluctuations of the planar flames in Section 2.4) and that the influence of flame stretch is negligible for obtaining the leading order result.

Attempts to explain the flame TF behavior have lead to the conclusion that the assumption of a uniform bulk oscillating flow contradicts the experimentally observed time delay behavior [110]. Because of this, attention recently shifted to the peculiarities of the oscillating flow upstream of the flame cone. It has been observed that the flow field at the burner outlet consists of periodical perturbations of a convective nature superimposed on the acoustic wave. These convective traveling waves inside the flame cone have been characterized experimentally [60,5,7] and numerically [63,94], but currently, it is not clear what causes these waves. It can be either the mean effect of the pressure change due to gas expansion (acceleration) at the flame front, or the effect of the time dependent (oscillating) flame retroaction to the upstream flow. Regarding the last possibility, two hypotheses are formulated. The first one states that the curved flame front has an effect on the flow upstream of the flame [7]. The second hypothesis states that the radial motion of the flame anchoring point drives the peristaltic motion [60]. Convective perturbations have also been observed in non-reacting free jets, as demonstrated in [8].

A first attempt to build a theoretical model describing the response of conical flames to an acoustic flow excitation was developed by Merk [88] on a semi-phenomenological basis. The proposed model leads to a first-order equation for the heat release rate and accordingly to low-pass filter features of the flame response to excitation. A manifest discrepancy between experiment and theory is that for the high frequency limit Merk's theory predicts a phase for the transfer function which is asymptotically approaching $\pi/2$. A next stage in the modeling was based on observed flame motions. Accordingly, theories developed in the 1970's prescribe the flame front perturbation as convective waves on the surface of the flame front, see, e.g., Matsui [82,120]. The results of these models suffer from the same shortcoming as Merk's model. A new stage in theoretical modeling was initiated by the idea to describe the kinematics of the excited flame front by the so-called G-equation [36]. The fundamentally new idea in this approach is that the flame front perturbations are not prescribed *a priori*, but naturally arise as the flame front responds to an applied flow perturbation. In early versions of the kinematic models, the flow perturbations were prescribed as bulk streamwise oscillations [36]. A more complete theory is presented in [108]. For the case of a conical flame and a one-dimensional acoustically perturbed velocity field given by $u(r,t) = \bar{u} + \hat{u}\cos(\omega t)$, $v(r,t) = 0$, with u the axial and v is the radial component, the expression for the flame area response TF is given by [108]:

$$\mathrm{TF}(\omega) = \frac{2}{\mathrm{Sr}^2}(1 + \mathrm{i}\,\mathrm{Sr} - \exp(\mathrm{i}\,\mathrm{Sr})), \tag{3.16}$$

with Sr the Strouhal number given by $\mathrm{Sr} = \omega R/(s_L \cos\beta) = (\omega h/\bar{u})(1 + (R/h)^2)$, where R and β are the burner radius and the half cone angle at the flame tip, respectively. The model correctly predicts the formation and further convection of flame front undulations but shows a phase saturation at $\pi/2$. If a convective wave is superimposed on the acoustic wave, i.e., $u = \bar{u} + \hat{u}\exp(\mathrm{i}(ky - \omega t))$ with $k = \omega/\bar{u}$, the flame TF does demonstrate the experimentally observed behavior typical for systems with a time lag [108]:

$$\mathrm{TF}(\omega) = \frac{2}{\mathrm{Sr}^2} \times \frac{1}{1-\cos^2\beta}\left(1 + \mathrm{i}\,\mathrm{Sr} - \frac{\exp(\mathrm{i}\,\mathrm{Sr}\cos^2\beta) - 1}{\cos^2\beta}\right). \tag{3.17}$$

The kinematical flame front tracking method allows predicting a perturbed flame form close to the experimentally measured one even in the case of cusped flames [6]. However, this theoretical TF does not contain the experimentally observed 'offset' part, which is relatively small and has a weaker frequency dependence. There is evidence that the TF offset originates from the flame foot region and is probably associated with the motion of the flame anchoring point, which is not considered in the current theory. In this zone the role of heat transfer between the flame and burner surface plays a dominant role and there is only one first provisional model which includes the effect of an oscillating heat loss to the burner surface [4].

The observed gain overshoot above 1 for a certain frequency range was theoretically described in the framework of kinematic models for V-flames [108], and is in accordance with experimental observations. The cause of the amplification of the response over its quasi-steady value lies in the interference between perturbations which originate from the anchoring point and travel along the flame [100]. A TF gain above 1 was also measured for multiple flame systems [62,63,92] and for other geometries including conical flames and M-shaped flames [31], but to our knowledge there are no models of flame thermo-acoustics that include these flame–flame interactions.

A next logical step in the development of an analytical model would be to predict the flow dynamics starting from the prescribed plane acoustic field at some position upstream of the burner. This step will probably require the incorporation of the flame retroaction on the upstream flow. Other necessary extensions of the analytical models should include (i) intrinsic mechanisms of flame anchoring to the burner which accounts for heat transfer to the burner surface and (ii) mutual interaction between adjacent flames. It is questionable whether these extensions are possible. Otherwise, one has to rely on numerical modeling.

Existing numerical models of the flame forced response can be classified in two groups. The first is based on the numerical solution of the G-equation where the upstream flow field is given and assumed to be unaffected by the flame motion, and the second applies direct numerical simulation to the full set of conservation laws for reacting flow. To correctly reproduce the flame sheet kinematics with the G-equation model the flow field should be prescribed *a priori*. However, the "natural" flame anchoring to the burner is an extra difficulty of the model. Therefore, the predictive ability of these flame tracking models is limited, albeit they are computationally efficient. Schuller et al. [107] show, however, that a direct integration of the G-equation combined with a suitable description of the incident velocity perturbation field yields adequate representations of the flame front kinematics and provides an accurate determination of the flame transfer function. On the other hand, direct numerical simulation of reacting flow is computationally expensive, but the results agree in general very well with experimental results. Simulations in [33,54,94,126] use compressible codes to reproduce the forced dynamics of conical flames. In [63] computations based on incompressible fluid dynamics with one step chemical reaction and a relatively simple molecular transport model were used to simulate the slit flame response to an acoustic perturbation. This investigation shows that the numerical model is capable to reproduce all aspects observed in the experiments. The correspondence of numerical simulation and experiments gives insight into the physics of flame–acoustics interaction, necessary to formulate improved analytical models. First, to predict the time lag it is necessary to combine the kinematic model with a model for the creation of convective waves at the burner outlet. Second, to predict the TF saturation, the jagged behavior and, probably the effect of the TF gain overshoot, it is necessary to develop a method to describe the dynamics of the flame anchoring zone. Finally, inclusion of the flame retroaction to the upstream flow and flame-to-flame interaction in the multiple flame configurations is probably required.

An interesting branch of research on forced flame dynamics has popped up recently. The idea is to use the knowledge obtained to control the flame thermo-acoustic behavior of a combustion system via proper modifications of the burner. One idea is the modification of the flow field perturbation by introducing obstacles in the channel (hole) in such a way that the traveling waves can be controlled, and accordingly, the time delay of the flame TF can be adjusted [56]. Another idea how to design an acoustically stable combustor is to combine different perforations in one burner deck. At the heart of this design approach lies the fact that the overall acoustic TF of the burner is a weighted superposition of TF's for different groups of holes and slits [62]. Such TF control methods bring the construction of the most desired TF within reach. However, for prescribing the desired TF behavior, an acoustic analysis of the complete combustor system is needed.

3.5. Body forces for flame control

Analogous to the case of one-dimensional flames, apart from heat transfer, centrifugal (gravitation) and electrical fields are sometimes applied to control Bunsen-type flames. Two control purposes are discussed in the literature: (i) flame anchoring on the burner to extend or shift the flashback and blow-off limits and (ii) control of dynamic (e.g., acoustic) instabilities. Flame control via centrifugal forces is technically difficult. Nevertheless, the rapid axial rotation of a burner tube has been investigated experimentally to observe changes in the stabilization of methane–air Bunsen flames [23]. Both flame blow-off and flashback limits have been determined as a function of the equivalence ratio and flow velocity. It is found that rotation of the system affects both stabilization limits. Research on the influence of electric fields on premixed flames has a long history. Various effects of an electric field (EF) on premixed flames are observed, and results were summarized in several books and reviews (see for instance [35,71,131]). Some observed effects related to Bunsen-type flame control are the following: (i) in a transversal EF (electrode plates parallel to the flame) the flame is inclined towards the negative electrode, (ii) in a streamwise directed

EF (the burner is one electrode and a mesh or ring above the burner acts as the other electrode) the flame is pushed towards the burner if the burner plate is the kathode; if the burner serves as anode, then at relatively low voltage the flame tends to blow-off from the burner but at high voltage the flame once again is pushed towards the burner leading to an increase of the flame blow-off velocity. The flame form is modified and in some cases the thermo-diffusive instability can be promoted by the EF leading to severe modification of the flame cone, see for instance [135]. Another aspect of flame control via an EF is related to the idea to alter the thermo-acoustic behavior of the flame. One of the first attempts to suppress thermo-acoustic oscillations in combustion was described by Abrukov et al. [1]. The effect of a DC EF on the heat release rate was attributed to the action of the ionic wind through a decrease in flame surface area. An idea to use an AC EF to control the acoustic instability was also considered. One way of this type of control is to measure some "indicator" of instability (for example, pressure fluctuation) and according to some control strategy apply a corresponding AC EF to the flame. In the simplest form (phase shift control) this idea was first applied by Weinaas and McChesney [130].

Joule heating is another effect of an EF which can be used for self-adaptive control of thermo-acoustic instabilities [2]. The idea follows from the fact that changes in flame heat release rate lead to in-phase changes in the conductivity of the combustion zone. If a constant current discharge is imposed on the combustion zone, then the variable component of the Joule heat release will be out of phase with the oscillations in the flame heat release. Therefore a substantial reduction in the fluctuating heat release can be expected. This technique was successfully demonstrated for the case of a premixed propane–air flame. The above-mentioned ideas, as well as some other attempts to use an electric field to control thermo-acoustic instabilities of different types of flames, are presented in a recently published book [3].

4. Conclusions and outlook

Current state of knowledge of premixed flame propagation, dynamics and control is reviewed. A distinction is made between planar and multi-dimensional flames.

Knowledge on the flame propagation of planar flames is well established and physical models to describe their behavior are mature. Experimental data of burning velocities of many fuels and mixtures are available. These data form the basis to derive chemical kinetics reaction mechanisms. The following points summarize current status and future requirements:

- Extension of the work to other future (bio)-fuels has started, but more measurements of burning velocities of such future fuels and of (surrogate) mixtures would be welcome in the near future, since they are expected to play an important role in our future energy supply.
- Obtaining burning velocity data with a higher accuracy is within reach (e.g., using the heat-flux method) and together with more reliable data of Markstein numbers, these data would be very helpful to further improve chemical kinetics models. Data for Markstein numbers need to be supplemented with details at which position in the flame front they are obtained.
- Our physical understanding of pulsating flame phenomena and other oscillatory modes is still limited. The simple model, presented here to enhance our understanding of pulsations and flame dynamics under the influence of acoustics forcing, indicates that parameters like the Zeldovich number, heat-loss and stand-off distance play an important role. It remains to be investigated whether the interpretation deduced from this simple model is in line with other more detailed models. Extension of this model to non-unit Lewis numbers and to incorporate internal flame dynamics would be very welcome. More experimental data, e.g., with flames of other fuels and mixtures, could help the interpretation further as well.
- For the case of planar flames, it can be stated that their 'control' by heat transfer is well understood, although it is expected that even more sophisticated methods will be used in future research to control their behavior, for instance based on electric, magnetic and acoustic actuation or using other body forces.

Multi-dimensional flames are also well understood, although the state of knowledge is less mature. The following points summarize current status and future requirements:

- Flame stretch and flame curvature definitions have been extended successfully, taking into account the internal structure of the flame front. This was the basis for the creation of a new flame stretch theory, by which we can reach quantitative predictions of the effects of local distortions on the burning velocity, also in the case of strong stretch, like in the tip of a Bunsen flame or in case of strongly wrinkled turbulent flames. The formalism is also applicable to combustion systems described by multiple species transport and chemistry models.
- On the other hand, for a better physical understanding and a more easy confrontation of the results of this theory with experiments, it would be desirable to simplify the current formalism. Some ways how to do this or how to

extend or improve current models are indicated in the text. First attempts to simplify the formalism focused on temperature variations and on a single (lean-species) Lewis number. For lean mixtures this has already been successful, but more simplifications are within reach. Stoichiometric and rich flames are less well understood and should be studied more deeply in the future.

- Current models used for the modeling of premixed flame propagation in complex environments (like based on the kinematic G-equation) mostly rely on older and less accurate flame stretch models and might need revision, especially in case of stoichiometric and rich mixtures. Models, based on the new flame stretch formalism are needed to predict small-scale and large-scale dynamics of premixed flames in a more quantitative way.
- Our understanding of the interaction of premixed Bunsen-type flames with acoustic waves has increased significantly in the last years. The characteristic response of such flames in terms of the flame TF is well established and has the form of a low-pass filter in frequency domain, combined with a constant time-delay phenomenon describing the TF phase, due to a convective wave propagating from the burner to the flame, and a saturation effect for high frequencies.
- Kinematic models for describing the underlying motion of these flames are well established, but the current kinematic models still have some serious shortcomings. The measured flame TF phase cannot be predicted, because of its lacking abilities to model flame foot motion on top of the burner and the flame retro-action on the unburned gas flow. Extensions of these models to account for flame–flow interaction, flame–flame interaction and flame–burner interaction are thus needed to solve these discrepancies. On the other hand, numerical modeling results do reproduce experimental observations.
- Finally, a multiple set of measures to adapt premixed flames to the combustion system of interest is at hand and different ways to influence these flames by heat transfer, acoustic waves and other less often used control methods like electric, magnetic and acoustic fields are available. This opens the way to modify and control premixed laminar flames in practical devices to guarantee safe and proper operation.

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